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A privacy-enhanced framework for collaborative Big Data analysis in healthcare using adaptive federated learning aggregation

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Abstract

The exponential growth of Big Data in healthcare, particularly in AI-driven medical diagnostics, has raised critical concerns about data privacy in medical image classification. With over 30% of healthcare organizations worldwide experiencing data breaches in the past year, the demand for secure, privacy-preserving solutions is more urgent than ever. This study explores a federated learning approach combined with transfer learning to enhance privacy in medical image classification using ResNet and VGG16 architectures. Pre-trained on ImageNet and fine tuned on three specialized medical datasets TB chest X-rays, brain tumor MRI scans, and diabetic retinopathy images these models were deployed in a simulated multi-center healthcare environment. A major contribution of this work is the development of an adaptive aggregation methodology, which dynamically selects between Federated Averaging (FedAvg) and Federated Stochastic Gradient Descent (FedSGD) based on real-time data divergence observed across participating clients. Unlike conventional static aggregation methods, which uniformly apply the same update rule regardless of data heterogeneity, the proposed adaptive approach monitors gradients and data distributions at each communication round and selects the most suitable aggregation method dynamically. This adaptive strategy not only improves convergence but also optimizes resource utilization, making it suitable for multi-center healthcare networks where data heterogeneity is prevalent. The novelty of the proposed adaptive aggregation lies in its ability to maintain robust performance while minimizing computational costs, making it feasible for large-scale healthcare AI networks, such as hospital federated learning systems. Comparative analysis with baseline FL models, including FedAvg and FedSGD, shows that the adaptive aggregation method achieves comparable accuracy (up to 96.3%) while significantly reducing execution time by approximately 20% and maintaining a competitive F1-score. Additionally, the integration of privacy-preserving techniques ensures that sensitive patient data remains secure throughout the learning process. By integrating transfer learning with federated learning, this study presents a scalable and privacy-preserving framework for Big Data analytics in healthcare. The findings underscore the potential of adaptive aggregation to enhance federated learning efficiency across heterogeneous datasets, enabling medical institutions to develop high-accuracy diagnostic models without direct access to patient data.

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Keywords: Federated learning, Big Data, Machine learning, Artificial intelligence, Data privacy, Transfer learning, Image classification, Deep learning

Introduction

The domain of medical imaging has experienced substantial growth in recent decades, propelled by advancements in artificial intelligence (AI), machine learning (ML), and deep learning (DL) [1–3]. Historically, medical image analysis depended on manual interpretation by radiologists and doctors, a labour-intensive procedure prone to variations in competence and human error. The merging of AI and ML techniques has altered medical image processing, providing more exact, consistent, and automated solutions for disease diagnosis and prediction [4, 5]. Deep learning has transformed medical imaging by allowing models to autonomously learn complex features from raw image data, eliminating the necessity for manual feature extraction and significantly enhancing the accuracy and efficiency of medical diagnostics [6, 7].

Artificial Intelligence, Machine Learning, and Deep Learning have become essential instruments in the identification and examination of both communicable and non-communicable diseases [8]. Communicable illnesses such as tuberculosis (TB), a significant global health concern, have experienced substantial advancements in early identification and treatment planning owing to AI-driven picture categorisation approaches. Chest X-rays [9, 10], a fundamental diagnostic instrument for tuberculosis, are now evaluated using convolutional neural network (CNN) models [11, 12], which may identify even subtle indications of the disease with greater precision than conventional techniques. This is essential in addressing tuberculosis, especially in resource-constrained environments, where automated AI solutions can assist overwhelmed healthcare systems. Research indicates that AI-driven diagnostic tools can attain sensitivity and specificity on par with professional radiologists, hence improving early detection efforts and bolstering worldwide tuberculosis eradication operations.

Likewise, for non-communicable disorders such as brain tumours and diabetic retinopathy, deep learning models have significantly enhanced medical picture processing [13, 14]. Brain tumours, typically identified via MRI scans, necessitate accurate segmentation and classification for optimal treatment planning. Deep learning methodologies, especially CNNs [15], have exhibited remarkable efficacy in differentiating between benign and malignant tumours, enhancing survival rates through earlier and more precise diagnosis. Deep learning models are currently extensively utilised to analyse retinal pictures and identify early-stage microaneurysms or haemorrhages in diabetic retinopathy, a prevalent eye condition and a leading cause of global blindness. AI-assisted technologies have enhanced the accessibility of diabetic retinopathy screening, particularly in underprivileged regions, while automated methods have alleviated the diagnostic workload for ophthalmologists, facilitating large-scale screening initiatives.

The integration of AI, ML, and DL in medical image processing has substantially enhanced diagnostic precision [16], lowered healthcare expenses, and broadened access to quality medical services. The swift integration of these technologies in the analysis of both communicable diseases and non-communicable diseases. This research expands on the increasing body of evidence illustrating the effectiveness of AI models in

improving medical image classification, while emphasising privacy-preserving methods via federated learning [17, 18]. This study seeks to enhance the application of AI in medical diagnostics by tackling the issues of privacy, accuracy, and scalability, thereby contributing to global healthcare improvement efforts. The involved architecture of federated learning can be seen in Fig. 1.

This research selected ResNet and VGG16 as the primary models for medical image classification via transfer learning due to their well-documented efficacy, robustness, and outstanding performance across a variety of medical datasets [19]. Both models were pre-trained on large-scale datasets such as ImageNet [20], enabling efficient fine-tuning on specialized medical image datasets, including TB chest X-rays, brain tumor MRI scans, and diabetic retinopathy images. ResNet's ability to perform multi-scale feature extraction and VGG16's deeper architecture allowed these models to capture intricate patterns in medical images, making them well-suited for detecting delicate features crucial for accurate diagnosis. While AlexNet was considered, its shallower architecture and lower parameter count were deemed insufficient for the complexity of medical image classification in this context, particularly when compared to the superior pattern recognition capabilities offered by ResNet and VGG16. Thus, these models were selected for their capacity to deliver higher classification accuracy and robustness, which are indispensable for medical diagnostics.

This study introduces a novel approach by employing adaptive aggregation in a federated learning framework as shown in Fig. 2, specifically designed for a multi-medical center image research model. This model enables multiple medical centers to leverage each other's learning without ever sharing sensitive patient data, ensuring privacy while enhancing collaborative learning. The proposed method dynamically switches between FedAvg [21] and FedSGD [22] based on the observed data divergence during communication rounds. This adaptive technique optimizes model convergence, allowing the models to generalize more effectively in heterogeneous data settings. The approach is particularly beneficial when utilizing pre-trained models like ResNet and VGG16, as it enhances the efficacy of transfer learning in distributed environments

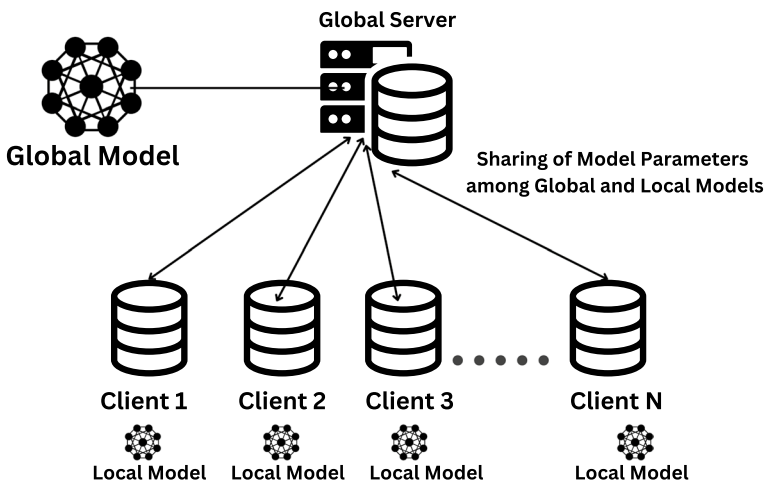


Fig. 1 Federated learning architecture

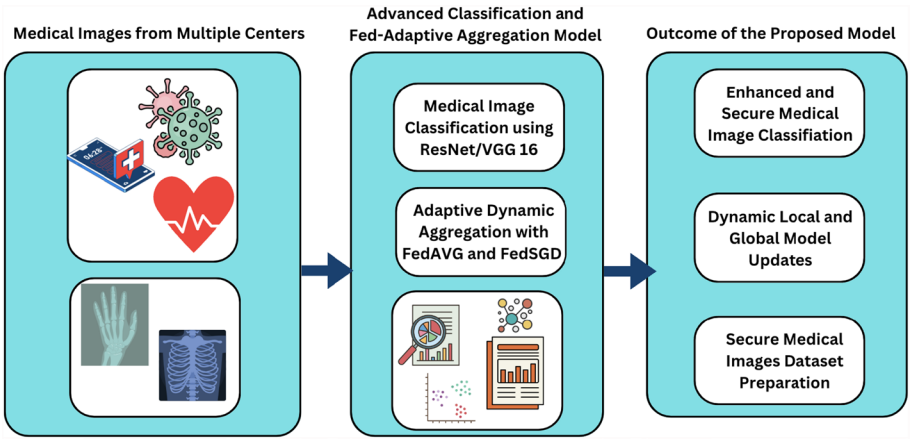


Fig. 2 Proposed adaptive aggregation FL framework

while preserving the privacy benefits of federated learning [23]. This novel aggregation method ensures that the federated learning framework adapts to the variability of the data across clients, significantly improving the convergence speed and accuracy of the models in privacy-sensitive, decentralized medical applications.

The study performs a comprehensive comparison of the performance of ResNet and VGG16, fine-tuned via transfer learning [24], in both centralized and federated settings. In the centralized setting, differential privacy was implemented to safeguard sensitive data during model training, while federated learning protocols were employed in the decentralized setting to maintain privacy without centralizing the data. This comparison highlights the trade-offs between accuracy [25], privacy, and computational efficiency, providing valuable insights into the performance of models in privacy-sensitive medical image classification tasks. Additionally, the study demonstrates how transfer learning, when integrated with federated learning, retains high levels of accuracy with minimal performance degradation typically less than 5% compared to centralized models reinforcing the feasibility of this approach for real-world medical diagnostics.

By introducing adaptive aggregation, this study advances the federated learning paradigm, providing a flexible and scalable approach for optimizing convergence in distributed medical image classification on large-scale healthcare datasets. This innovation significantly enhances the potential of privacy-preserving AI solutions in medical diagnostics, empowering healthcare institutions to train highly accurate models while upholding stringent data privacy standards. The adaptive aggregation method improves both the efficiency and scalability of the federated learning framework, ensuring robust generalization across diverse and expansive medical datasets. This approach makes collaborative learning across multiple medical centers not only more secure but also better suited to the demands of Big Data in healthcare.

FL has emerged as a promising solution for privacy-preserving medical AI applications, yet it faces critical challenges in real-world deployments. One of the most significant issues is heterogeneity in medical datasets, where data distributions across different healthcare institutions are highly non-IID, leading to imbalanced model updates and degraded performance. Additionally, computational constraints in federated environments pose challenges, as participating clients may have varying

processing power, bandwidth, and storage capabilities, affecting training efficiency and scalability. The main and unique contribution of this study addresses these challenges by introducing an adaptive aggregation strategy that dynamically switches between FedAvg and FedSGD based on real-time data divergence. Unlike traditional static aggregation methods, this approach enhances model convergence, optimizes learning efficiency, and mitigates performance gaps in heterogeneous settings. Furthermore, this work integrates transfer learning with federated learning by leveraging pre-trained CNN models, reducing on-device training requirements while preserving classification accuracy. Through extensive evaluation on TB chest X-rays, brain tumor MRI, and diabetic retinopathy datasets, the study demonstrates the scalability and robustness of the proposed framework in multi-center medical environments. These contributions distinguish this work from prior studies, providing an optimized, privacy-preserving, and computationally efficient federated learning approach tailored for real-world healthcare applications.

Conventional FL aggregation methods, such as FedAvg, struggle to maintain optimal performance when applied to heterogeneous medical datasets. This is primarily because FedAvg aggregates client models by averaging their updates without accounting for data divergence. In medical scenarios, where data collected from different institutions can vary significantly in terms of patient demographics, imaging protocols, and disease prevalence, this uniform aggregation strategy leads to slower convergence and suboptimal model performance. The proposed adaptive aggregation addresses this limitation by dynamically selecting the most suitable aggregation technique (FedAvg or FedSGD) based on the observed divergence, thus improving convergence speed and accuracy in non-IID settings.

The advantages that the Adaptive Aggregation method has over the other conventional methods can be seen from the Table 1.

Selection of specific pre-trained models VGG16 and ResNet

Federated learning in medical imaging requires architectures that can generalize across varying computational environments, data distributions, and model complexities. To ensure broad applicability, this study employs ResNet and VGG16, representing two distinct categories of convolutional neural networks CNNs. VGG16 is a classical deep learning architecture known for its sequential convolutional layers, making it

Table 1 Comparison of adaptive aggregation with privacy-enhanced federated learning methods

Aggregation method	Privacy technique	Efficiency	Performance on non-IID data	Computational overhead
Adaptive Aggregation (FedAvg + FedSGD)	Real-time data divergence monitoring	High	High	Moderate
Differential Privacy-based FL	Noise addition to gradients	Moderate	Low	High
Homomorphic Encryption-based FL	Encrypted model aggregation	Low	Low	Very high
Static FedAvg	Parameter averaging	High	Low	Low
Static FedSGD	Gradient aggregation	Moderate	Moderate	High

computationally expensive but widely used in medical image classification tasks [26]. On the other hand, ResNet introduces residual learning, enabling deeper networks without suffering from vanishing gradient issues, making it a robust and efficient architecture for feature extraction [27]. The selection of these architectures ensures that the proposed adaptive aggregation approach remains effective across both heavy, classical models and modern, lightweight architectures.

Alternative architectures such as EfficientNet and DenseNet have demonstrated superior parameter efficiency and reduced computational overhead in various cases depending on the divergence and dataset complexity [28, 29]. However, the objective of this study is to assess the generalisability of adaptive aggregation across diverse model complexities. The inclusion of EfficientNet and DenseNet would have limited the study to modern architectures alone, neglecting classical CNN architectures that are still widely utilized in real-world applications. By selecting ResNet and VGG16, the study effectively evaluates the adaptability of federated learning across a spectrum of model complexities, from conventional deep networks to modern residual-based frameworks.

Fine-tuning vs. Training from scratch in federated learning

Training deep learning models from scratch requires extensive computational resources and large-scale labeled datasets, both of which are often constrained in federated learning environments. Consequently, fine-tuning pre-trained models has emerged as a practical alternative, significantly improving convergence efficiency and generalization [30, 31]. In this study, ResNet and VGG16 were fine-tuned on medical imaging datasets, leveraging knowledge transfer from pre-trained weights while adapting to specific domain characteristics. This approach not only accelerates training convergence but also reduces the risk of overfitting, a crucial consideration given the limited size and privacy constraints of federated medical datasets.

Prior research has established that training CNNs from scratch is computationally prohibitive in distributed learning setups [32]. Given the decentralized nature of federated learning, where edge devices operate under limited processing power and communication bandwidth, fine-tuning pre-trained models ensures optimal performance without excessive resource consumption. The choice of fine-tuning over training from scratch aligns with existing best practices in federated learning, allowing the adaptive aggregation approach to remain computationally feasible while maintaining high classification accuracy.

This discussion underscores the necessity of selecting appropriate architectures and training methodologies that align with the computational constraints and privacy-preserving objectives of federated learning. The integration of both classical and modern CNN architectures, along with fine-tuning rather than full training, ensures that the proposed adaptive aggregation mechanism remains widely applicable across diverse real-world scenarios.

The remainder of this paper is organized as follows. Section Related work presents an overview of the related work and background studies relevant to this research, including previous developments in federated learning and privacy-preserving medical data mining. Section Data and methodology elaborates on the methodology employed in this study, detailing the datasets utilized and the novel model introduced for collaborative

medical image classification using adaptive federated aggregation techniques. This section highlights the unique aspects of the federated learning framework implemented across multiple medical centers. Section Results discusses the results of the experiments, providing a comprehensive analysis of the key findings, including model performance under different settings and the impact of the proposed adaptive aggregation strategy. Finally, Section Policy Implications and Potential Real WorldDeployment concludes the article by summarizing the key contributions, addressing limitations of the study, and outlining potential future research directions that could further enhance federated learning in medical data mining and classification tasks.

Major contributions

In this study, the following major contributions have been made to enhance the application of federated learning FL in privacy-preserving medical image classification:

1. *Novel Adaptive aggregation method* The proposed adaptive aggregation method dynamically switches between FedAvg and FedSGD based on the divergence observed across client updates. Unlike conventional aggregation techniques, this approach ensures faster convergence and improved accuracy by optimizing the aggregation strategy in real-time according to the data distribution heterogeneity.
2. *Enhanced privacy in healthcare AI* The integration of adaptive aggregation within the federated learning framework significantly enhances data privacy by maintaining decentralized model training. This approach allows medical institutions to collaboratively train models without direct access to patient data, thereby reducing the risk of data breaches while preserving model utility.
3. *Scalability and robustness* By dynamically adapting to varying levels of data divergence, the proposed method demonstrates robustness against data heterogeneity and exhibits scalable performance in large multi-center healthcare environments. This flexibility makes the proposed framework highly applicable in real-world settings.

Related work

See Table 2.

Jin et al. [38] offer a co-aware personalised federated learning model that integrates metadata and picture information for intelligent medical applications. The system incorporates both metadata and picture elements to improve diagnostic accuracy. A tailored federated learning technique utilising parameterised knowledge transfer is developed to enhance model performance by acquiring insights from edge nodes with greater contributions. A Naïve Bayes classifier is employed to categorise metadata, and the outcomes obtained from both metadata and image classifiers are consolidated to enhance diagnostic precision. The model's customised aggregation process surpasses conventional federated learning methods by accounting for the individual contributions of all clients to the master model. The method has an accuracy of 97.16% on the PAD-UFES-20 dataset. The model is constrained by its dependence on merely two data

Table 2 Comparison of Related Work in FL-based Healthcare Systems

Reference	Local model algorithm (ML model)	Global model algorithm (FL model)	Key contribution	Limitation
Abou El Houda et al. [33]	Gradient Descent (GD), CNN (for image classification)	Ring-AllReduce FL with threshold secret sharing	Introduced a decentralized FL model for robust and privacy-preserving gradient aggregation. Enhanced resilience to client dropouts using secret sharing.	Communication overhead due to lack of optimization for model aggregation in large-scale settings.
Ullah et al. [34]	CNN (for COVID-19 X-ray classification)	Federated Averaging (FedAvg)	Applied FL to COVID-19 diagnosis, improving classification accuracy across distributed clients (hospitals) without centralized data sharing.	Performance decreases with fewer clients, and the model's adaptability to other diseases was not thoroughly evaluated.
Abaoud et al. [35]	Logistic Regression, Decision Trees	Secure Multi-party Computation (SMPC) based FL	Proposed a privacy-preserving FL model with local differential privacy and secure computation, focusing on scalability and security.	Increased computation time due to secure computation techniques, and scalability to larger datasets was not fully tested.
Shiranthik et al. [36]	Multi-layer Perceptron (MLP)	Decentralized Ring-based FL	Focused on reducing communication overhead in FL systems, using a ring-based model to ensure better scalability for large-scale applications.	The approach is less effective for models with more complex architectures like CNNs, and privacy concerns were not fully addressed.
Kalapaaking et al. [37]	Deep Neural Network (DNN) with two hidden layers	SMPC-based FL combined with Blockchain	Introduced HealthFed, integrating blockchain with FL to ensure decentralized privacy-preserving collaboration among healthcare institutions. Tested on Breast Cancer Wisconsin dataset.	High complexity due to blockchain integration; computational costs increase with more collaborators.
Haripriya et al. [This Study]	ResNet and VGG-16 for multiple healthcare image datasets	Novel Adaptive Aggregation FedAVG+FedSGD	Introduced adaptive aggregation as a novel approach to federated learning, combining the strengths of both FedAvg and FedSGD with dynamic aggregation on the basis of convergence.	Computational overhead was significant for larger models like VGG-16, indicating the need for more optimized architectures or compression techniques in future work.

modalities, yet there exists an opportunity for enhancement by the integration of supplementary modalities, such as video or more intricate patient data.

Ali et al. [39] presents a confidentiality-preserving federated learning paradigm for medical picture classification, emphasising the protection of patient privacy while maintaining high accuracy in medical diagnostics. The FL model utilises CNNs for image processing and secure multi-party computing (SMPC) methods to protect sensitive medical data during distributed model training. The findings indicate that the model attains comparable accuracy to centralised approaches while upholding stringent privacy restrictions. Notwithstanding its advantages, the framework's significant dependence on privacy techniques like SMPC escalates computational expenses, hence constraining its scalability in resource-limited healthcare settings. Furthermore, its capacity to process intricate or non-image-based healthcare information was not thoroughly investigated.

Tiwari et al. [40] examines the utilisation of federated reinforcement learning (FRL) inside consumer-oriented Internet of Medical Things (IoMT) systems for cyborg applications. The proposed system combines multi-agent federated reinforcement learning protocols (MFRLP) with socket-based calls for remotely performed procedures (RPC) to enhance cyborg-enabled healthcare services. The authors employ a synthesis of reinforcement learning agents and deep Q-learning networks (DQN) to address dynamic environmental changes. Training and verifying the models across fog nodes substantially decreases waiting times and computational overhead. The framework accomplished a 50% decrease in training and testing duration and a 40% decrease in local processing time relative to conventional machine learning models. The paper recognises limits regarding security, especially for independent nodes, and proposes that future research might include blockchain to address these concerns.

Tanim et al. [41] introduces a privacy-preserving federated learning architecture tailored for cyborg-based applications in healthcare, particularly focused on knee surgery procedures. The methodology integrates deep reinforcement learning (DRL) with federated learning to analyse medical data in fog computing settings. The system delegates duties through mobile agents to fog nodes through a hierarchical learning model, ensuring optimal data processing and minimising communication delays. A significant contribution is the implementation of dynamic scheduling to equilibrate processing loads between mobile and server agents. The model exhibits significant enhancements in processing speed and data protection. The authors acknowledge that data leaks and resource inefficiencies persist as issues, highlighting possible areas for further optimisation in future implementations.

Alzubi et al. [42] examines the implementation of federated learning in conjunction with secure multi-party computation (SMPC) for decentralised healthcare solutions. It underscores the minimisation of communication latency in mobile healthcare settings while safeguarding data privacy via SMPC. The research emphasises an adaptive scheduling technique designed to enhance work allocation across edge devices and central servers, hence boosting execution time and data security. The approach provides substantial benefits in minimising latency and improving data privacy; but, it encounters challenges in managing larger datasets and more intricate healthcare situations, which may affect its effectiveness in practical implementations.

Nguyen et al. [43] provides a comprehensive assessment of the application of FLin the healthcare industry, focusing on its potential to improve privacy and scalability in dispersed healthcare systems. FL is broken down into a number of different types, such as horizontal, vertical, and transfer learning, and the authors examine its application in a number of important domains, such as medical imaging, remote health monitoring, and health data management. The paper highlights multiple drawbacks, including the dependence on non-healthcare datasets for assessing models, neglected privacy risks, and interaction and computational expenses in resource-constrained environments. Despite the fact that facial recognition technology has the potential to protect individuals' privacy and make personalized healthcare more accessible, the paper emphasizes several limitations. In the publication, the importance of conducting additional research is emphasized, particularly with regard to the application of FL to healthcare-specific datasets and the creation of more robust strategies for enhancing privacy. In general, it highlights the promise of FL while also addressing the problems that need to be solved in order to fully comprehend the benefits that it offers in the field of smart healthcare.

Metaheuristic algorithms inspired by natural and strategic behaviors have been increasingly applied to federated learning optimization. One such approach is the Greylag Goose Optimization (GGO) algorithm by El-Kenawy et al. [44], a swarm-based optimization method that mimics the aerodynamic efficiency of geese flying in a "V" formation. This algorithm optimizes global model updates by improving gradient-based communication efficiency, reducing computational overhead, and maintaining balanced weight adjustments in a decentralized setting. The use of swarm intelligence in federated learning could lead to more adaptive aggregation strategies, particularly in environments where client participation is heterogeneous or unstable.

Similarly, the Football Optimization Algorithm (FbOA) by El-Kenawy et al. [45] introduces a population-based metaheuristic approach inspired by football team dynamics. By balancing exploration and exploitation, this technique allows for strategic model updates that enhance federated learning performance, particularly in high-dimensional and non-linear problem spaces. The algorithm's tactical positioning mechanisms could be leveraged to dynamically adjust learning rates and participation strategies in federated learning environments, mitigating challenges associated with non-IID data distributions and slow convergence rates.

The survey by Umirzakova et al. [46] underscores the importance of accurate medical image restoration in enhancing the performance of deep learning models in healthcare. Low-resolution images, often generated by IoT devices, introduce biases that impact clinical decision-making. While super-resolution techniques address these challenges, achieving optimal restoration remains critical. Federated learning, augmented with blockchain for secure and decentralized collaboration, offers a promising framework to improve image restoration accuracy, enhance model training outcomes, and ensure trust and privacy in multi-center medical environments. This integration aligns with the need for scalable and reliable AI solutions in healthcare.

Khafaga et al. [47] presented a novel approach to automate the diagnosis of monkeypox disease by leveraging deep convolutional neural networks (CNNs) fine-tuned using the Al-Biruni Earth radius (BER) optimization-based stochastic fractal search (BERSFS).

This method aimed to enhance the accuracy of monkeypox classification from low-resolution images, a task traditionally requiring high-level medical expertise. The study demonstrated that using optimized CNN layers and triplet loss functions significantly improved the classification of monkeypox cases from human skin disease images. Statistical tests, including Wilcoxon and ANOVA, confirmed the superior performance of the proposed model compared to conventional machine learning methods.

Abdullah et al. [48] focused on the challenges of traffic congestion prediction, proposing a bidirectional recurrent neural network (BRNN) with Gated Recurrent Units (GRUs) to classify traffic states in smart cities. By incorporating features like speed, weather, and accident probabilities, the model outperformed existing state-of-the-art approaches. The research highlights the importance of real-time data from sensors and connected devices, emphasizing the need for efficient congestion prediction models in urban environments. The study also explored future research directions, focusing on large-scale motion prediction accuracy and efficiency.

El-Kenawy et al. [49] explored the optimization of metamaterial antennas using machine learning techniques. The study proposed an optimal ensemble model for predicting bandwidth and gain by integrating machine learning algorithms, including Support Vector Machines (SVM), Random Forest, K-Neighbors Regressor, and Decision Tree Regressor. The authors employed the Adaptive Dynamic Polar Rose Guided Whale Optimization method (AD-PRS-Guided WOA) for feature selection. The results indicated that the proposed model, particularly using Random Forest, significantly improved prediction accuracy compared to traditional models.

Al-Mahdawi et al. [50] addressed the inverse initial value problem (IVP) for the heat equation, where the initial temperature distribution is unknown. The study proposed solving the problem by the separation of variables method while mitigating instability through the Picard method, which regularizes the operator mapping. The approach offered a stable approximation of the ill-posed problem, demonstrating that the inverse problem could be reliably addressed with controlled errors.

Towfek and Elkanzi [51] emphasized the role of AI and ML in combating infectious diseases, particularly in real-time disease surveillance and epidemic prediction. The study underscored the limitations of traditional epidemic models and advocated for integrating AI techniques to enhance the accuracy of forecasts and control measures. The authors discussed challenges such as data standardization, ethics, and transparency, highlighting the need for global collaboration to harness AI for public health interventions. The study also pointed to the potential of AI in advancing epidemic preparedness, leveraging genomic data and explainable AI to mitigate future pandemics.

Shamrat et al. [52] addressed the critical issue of early diagnosis of heart failure (HF) by developing a Multi-objective Stacked Enable Hybrid Model (MO-SEHM) that integrates the Non-dominated Sorting Genetic Algorithm II (NSGA-II) with a Stacked Enable Hybrid Model (SEHM). This approach aims to enhance feature selection while improving classification accuracy. The proposed MO-SEHM model, applied to a heart failure dataset from the Faisalabad Institute of Cardiology, demonstrated a high accuracy of 94.87% using nine relevant features. To ensure model transparency, Local Interpretable Model-agnostic Explanations (LIME) were used, making the system interpretable for both patients and stakeholders. Similarly, Shammi et al. [53] emphasized the role of

artificial intelligence (AI) in modern healthcare for chronic disease diagnosis, focusing on diseases like heart, diabetes, kidney, liver, and thyroid conditions. The study Shamrat et al. [54] reviewed AI's advancements in disease identification and highlighted the potential of integrating blockchain for secure healthcare data management. The decentralized and immutable characteristics of blockchain can enhance patient data security, promote interoperability, and give patients more control over their medical information.

In another study, Barai et al. [55] explored the automated classification of diabetic retinopathy (DR) stages using a novel CNN model named DRNet13, which achieved an accuracy of 97% on fundus image classification. The model outperformed other pre-trained architectures, demonstrating superior speed and efficiency. The study also employed various image preprocessing techniques, including noise reduction and image enhancement, to improve model performance. Banik et al. [56] introduced "RetiNet", a hybrid model combining ResNet50 and DenseNet121 features to diagnose retinal disorders like age-related macular degeneration and diabetic macular edema using optical coherence tomography (OCT) images. The model achieved a test accuracy of 98.50%, outperforming existing standards, and was implemented as a web application for diagnostic assistance. These studies underscore the growing importance of AI-driven models in medical diagnostics and emphasize the integration of explainability and data security to enhance practical implementation.

Recent advancements in federated learning have seen the integration of Vision Transformers and Self-Supervised Learning techniques to improve model performance and adaptability in decentralized environments. Li et al. [57] proposed a novel transformer-in-transformer facilitated FL framework that addresses challenges related to heterogeneous client data by introducing an adaptive-personalization mechanism. This approach departs from traditional methods by employing a hypernetwork on the server side, generating personalized projection matrices for the self-attention layers of each client. These matrices enhance the query, key, and value parameterization, significantly improving the model's scalability and efficiency. Experimental results demonstrated that the proposed framework outperforms conventional FL models, particularly in scenarios with non-IID data, highlighting the effectiveness of adaptive-personalization in maintaining high performance across diverse client distributions.

Similarly, Yan et al. [58] introduced a self-supervised learning framework for federated medical image analysis, leveraging transformer-based masked image modeling to improve robustness against heterogeneous data distributions. The framework enhances the representation learning of models in decentralized setups by directly performing self-supervised pre-training on decentralized target task datasets. This methodology facilitates effective knowledge transfer to downstream tasks while reducing reliance on large-scale labeled data. Empirical evaluations on retinal, dermatology, and chest X-ray classification tasks demonstrated significant improvements over traditional supervised FL models, with accuracy gains of up to 5.06% in non-IID settings. The study further highlighted that models pre-trained using the proposed SSL framework generalize better to out-of-distribution data and exhibit robust performance even when fine-tuned with limited labeled data, thereby validating the efficacy of transformer-based self-supervised learning in federated environments.

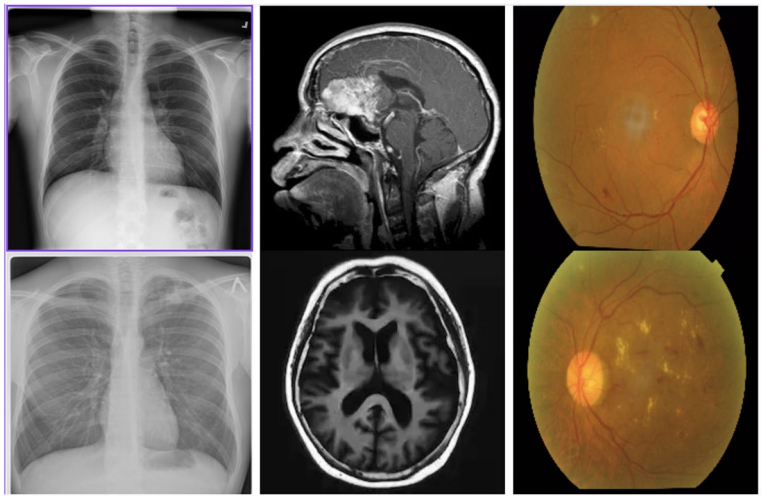


Fig. 3 Sample images of all the datasets utilized in this study

Table 3 Details of the medical image datasets used in the research

Dataset	Number of images	Categories	Image format	Image size (pixels)
TB Chest X-ray Dataset	7000	Normal X-rays, TB X-rays	JPEG	224 × 224
Diabetic Retinopathy Dataset	3662	No_DR, Mild, Moderate, Severe, Proliferate_DR	JPEG	224 × 224
Brain Tumor Dataset	3624	No Tumor, Meningioma, Glioma, Pituitary Tumor	JPEG	224 × 224

Data and methodology

Data and preprocessing

The datasets used in this study were sourced from publicly accessible platforms like Kaggle, providing a diverse collection of medical images essential for evaluating the scalability and effectiveness of the proposed models in a Big Data healthcare scenario. The TB chest X-ray dataset, for instance, consists of 7000 images-3500 normal and 3500 indicative of tuberculosis enabling the models to effectively distinguish between healthy and tuberculosis-affected cases. This dataset volume and diversity closely simulate the large-scale data environment often encountered in real-world healthcare analytics, offering a robust basis for testing the models' capability to perform accurate and privacy-preserving diagnostics at scale. Sample images from the dataset are shown in Fig. 3, illustrating the dataset's variability and its relevance in a Big Data context for healthcare.

The diabetic retinopathy dataset comprises 3,662 JPEG photos, classified into five categories: No DR, Mild, Moderate, Severe, and ProliferateDR, which denote different stages of diabetic retinopathy. This dataset facilitates the classification of various disease phases, enhancing early diagnosis and intervention options in diabetes patients.

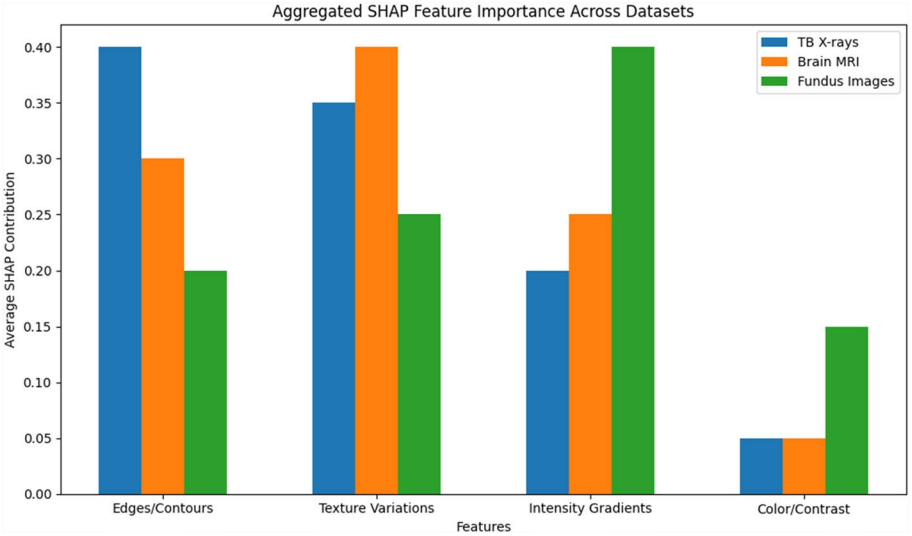


Fig. 4 Aggregated SHAP feature importance across datasets

Table 4 Summary of Key SHAP feature importance across datasets

Feature	TB X-rays	Brain MRI	Fundus Images	Explanation
Edges/contours	0.40	0.30	0.20	Crucial for identifying shape and structure differences, especially prominent in X-rays.
Texture variations	0.35	0.38	0.25	Important for detecting irregularities or abnormalities in tissue patterns. Highly relevant for MRI scans.
Intensity gradients	0.20	0.25	0.40	Helps detect variations in brightness and contrast, essential for fundus images where intensity differences indicate retinal changes.
Color/contrast	0.05	0.06	0.15	Relevant for fundus images as color variations can indicate vascular issues or retinal damage.

The brain tumour dataset comprises 3,624 JPEG photos categorised into four groups: no tumour, meningioma, glioma, and pituitary tumour. This dataset offers a variety of brain MRI images essential for tumour type classification, crucial for treatment planning and clinical decision-making.

To guarantee compatibility with the pre-trained models ResNet and VGG16 which employed in this study, all photos were preprocessed to conform to the input size specifications of these models, specifically scaling each image to 224x224 pixels as shown with the dataset details in the Table 3. This preprocessing procedure was crucial for preserving consistency among datasets and guaranteeing optimal model performance throughout training and evaluation.

To evaluate the significance of various features in the classification tasks, SHAP (SHapley Additive Explanations) analysis was conducted across all datasets, focusing on key image characteristics such as Edges/Contours, Texture Variations, Intensity Gradients, and Color/Contrast. These features were selected based on their relevance to the dataset-specific classification tasks. For instance, edges and contours were critical in identifying structural abnormalities in TB X-rays, while texture variations

and intensity gradients played a pivotal role in detecting tumors in Brain MRI and pathological changes in Fundus Images.

The results, as illustrated in Fig. 4, and Table 4 demonstrate the varying importance of these features across datasets. For TB X-rays, Edges/Contours contributed the most to the classification, reflecting their role in detecting structural abnormalities. In Brain MRI, Texture Variations were the dominant feature, aiding in the identification of irregular tissue patterns. For Fundus Images, Intensity Gradients and Color/Contrast were highly significant, highlighting areas indicative of diabetic retinopathy.

This analysis underscores the adaptability of the proposed model, which effectively leverages dataset-specific features, showcasing its generalizability across heterogeneous medical imaging tasks. The SHAP-based feature importance analysis provides valuable insights into the model's decision-making process, ensuring transparency and reliability.

Non-IID data handling and standardization

In practical healthcare applications, data collected from different hospitals often demonstrate significant variability. This experimental setup introduced extreme non-IID scenarios where 3 out of 10 clients (Client 2, Client 5, and Client 8) displayed substantial class imbalance, with one particular class being highly underrepresented. Such scenarios reflect real-world data biases in healthcare.

To address this, the Synthetic Minority Over-sampling Technique (SMOTE) was employes to mitigate data imbalance during the preprocessing stage. SMOTE effectively increases the representation of minority classes by generating synthetic samples, thereby reducing class bias while preserving data distribution characteristics. The SMOTE analysis showed that the model maintained robustness up to a certain level of infrastructural and data heterogeneity, as seen in the balanced client setup after preprocessing. Figure 5 illustrates the client data distribution before and after applying SMOTE. The left subplot shows the initial biased distribution where Clients 2, 5, and 8 had significantly lower data representation. The right subplot shows the balanced distribution after SMOTE preprocessing.

To ensure that the data distribution remains consistent after applying SMOTE, a Chi-square test was performed for uniformity. This test statistically validates that the processed data does not significantly deviate from a uniform distribution. The calculated

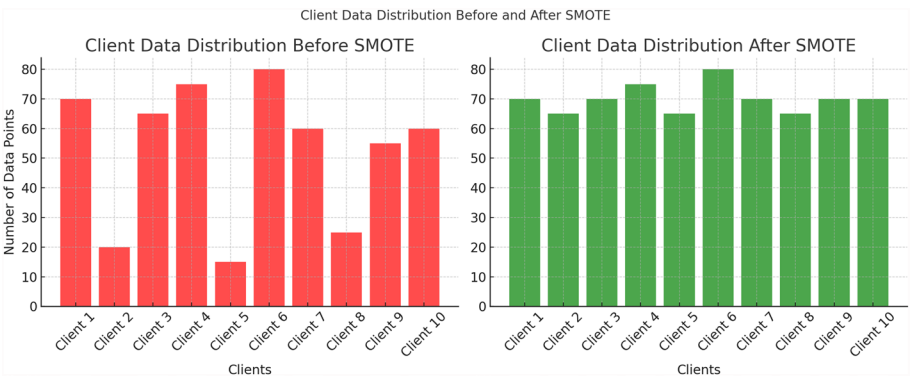


Fig. 5 Client data distribution before and after SMOTE

p-value ($p > 0.05$) indicates that the data, after balancing, does not exhibit a statistically significant bias, thereby making it well-suited for the adaptive aggregation process.

Operations

In this research, fine-tuning and validating the pre-trained models VGG16 and ResNet were central elements of the transfer learning approach applied for medical image classification. The datasets used in this study were partitioned into training, validation, and testing sets, with 70% of the data designated for training, 15% for validation, and 15% for testing. This equitable distribution was crucial to ensure that the models were sufficiently trained while maintaining an unbiased assessment during the validation and testing phases.

Fine-tuning involved adjusting the weights of VGG16 and ResNet, which were initially trained on ImageNet, to fit the specific medical image datasets used in this work. The foundational layers of both models, capturing essential features like edges and textures, were preserved to leverage their generalized feature extraction capabilities. In contrast, the higher layers were adapted to recognize domain-specific features relevant to medical imaging, such as identifying anomalies in TB chest X-rays, detecting brain tumor structures, and classifying stages of diabetic retinopathy. This process enabled rapid acquisition of medical-specific patterns without the need to train models from scratch, significantly reducing computational time and resource demands.

VGG16's deep architecture with multiple convolutional layers allows it to capture rich hierarchical features, which are beneficial for complex medical image analysis tasks. ResNet's residual connections help mitigate issues like vanishing gradients, allowing the network to train deeper layers effectively without loss of information, which is beneficial for complex medical image analysis tasks. These architectural advantages made VGG16 and ResNet especially suited for the diverse and intricate requirements of medical image classification in this study.

VGG16 and ResNet were chosen for this experiment over other pre-trained transfer learning models due to their unique architectural advantages, which make them particularly suited for handling complex medical image classification tasks in a federated setting. VGG16's deep architecture allows it to capture fine-grained details and hierarchical representations, which are crucial for tasks such as identifying subtle anomalies in diabetic retinopathy images or delineating tumor boundaries in MRI scans. ResNet, on the other hand, utilizes residual connections, allowing it to effectively train deeper networks without succumbing to vanishing gradients, which is advantageous for learning intricate and hierarchical representations in medical images.

These architectural strengths provide significant advantages over other models, such as Vgg19, which, while powerful, has a higher parameter count and lacks the residual learning mechanism, leading to potentially higher computational demands and increased risk of overfitting on small datasets. The comparison between Vgg16 and ResNet in this study offers a broad, practical perspective on balancing multi-scale feature extraction and deep learning efficiency, both of which are critical for collaborative and privacy-preserving analysis across multiple medical centers. This approach ensures that essential diagnostic features can be identified accurately while maintaining efficiency and privacy in a federated learning environment.

During training, both models were iteratively updated using backpropagation and gradient descent. The model parameters $\mathbf{w}$ were optimized by minimizing the categorical cross-entropy loss $\mathcal{L}$, defined for multi-class classification as:

$$\mathcal{L}(\mathbf{w}) = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c}) \tag{1}$$

where N is the number of training samples, C is the number of classes, $y_{i,c}$ is the true label of class c for sample i, and $\hat{y}_{i,c}$ is the predicted probability for class c. The gradients of the loss function with respect to the parameters were computed, and the model was updated via gradient descent:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta\nabla\mathcal{L}(\mathbf{w}^{(t)}) \tag{2}$$

where η is the learning rate, and $\nabla\mathcal{L}(\mathbf{w}^{(t)})$ represents the gradient of the loss function at iteration t.

To prevent overfitting, regularization techniques such as dropout were employed during training. In dropout, a fraction p of the neurons in each layer were randomly set to zero during each iteration, effectively reducing model complexity. Additionally, early stopping was implemented to halt training if the validation loss did not improve after a specified number of epochs, thus preventing overfitting.

The validation set was utilized to monitor the performance of the models during training, ensuring that the fine-tuning process for VGG-16 and ResNet did not result in overfitting or underfitting, ultimately enhancing the model's adaptability and robustness for medical image classification tasks.

VGG 16 model

The VGG16 model was utilized for medical image classification due to its depth and robust feature extraction capabilities [59]. VGG16, with its 16 layers as shown in Fig. 6, is well-suited for learning detailed hierarchical features in medical images such as TB chest

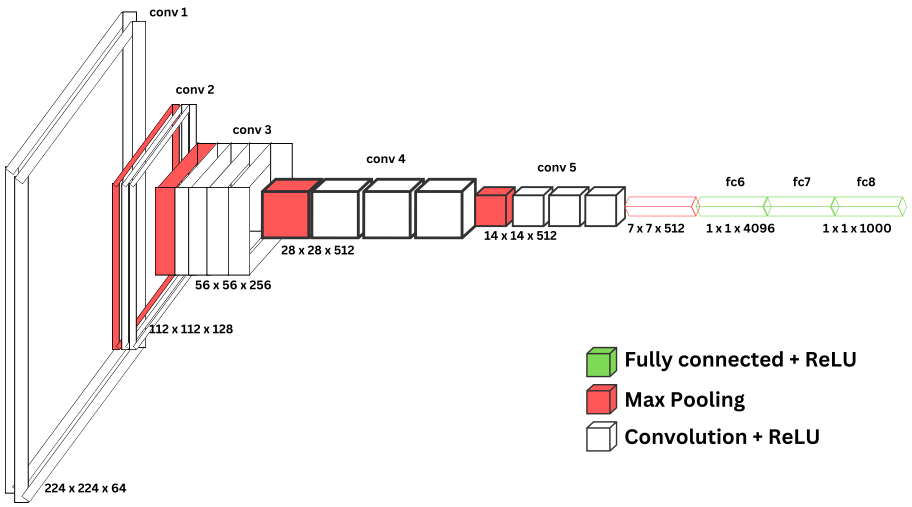


Fig. 6 VGG-16 Architecture

X-rays, brain tumor MRIs, and diabetic retinopathy images [60]. The convolutional layers in VGG16 use fixed-sized filters (3x3) to extract features from input images, which can be represented as:

$$\mathbf{y}_{i,j} = \sum_{m=1}^{M} \sum_{n=1}^{N} \mathbf{w}_{m,n} \mathbf{x}_{i+m,j+n} + b \tag{1}$$

where $\mathbf{x}$ represents the input image, $\mathbf{w}$ is the filter, b is the bias, and $\mathbf{y}$ is the output feature map. This systematic convolution over medical images allows VGG16 to capture both low-level and high-level features critical for medical diagnosis.

The suitability of VGG16 for this research lies in its ability to capture fine-grained details necessary for the accurate classification of medical conditions. The network's depth provides the advantage of learning complex patterns from medical images such as small lesions in diabetic retinopathy or subtle tumor regions in brain MRIs. Additionally, VGG16's architecture, when fine-tuned through transfer learning, was able to generalize well on the medical datasets after being trained on a large dataset like ImageNet, thus reducing the computational cost and training time.

ResNet model

ResNet was selected for medical image classification due to its deep learning architecture, as shown in Fig. 7, which incorporates residual connections to enable efficient training of very deep networks [61]. These residual connections, also known as skip connections, help to mitigate the vanishing gradient problem by allowing gradients to bypass multiple layers, making it feasible to train networks with hundreds of layers effectively [62, 63]. This approach allows ResNet to learn complex representations that capture both fine-grained and large-scale features within medical images, such as TB chest X-rays, brain MRIs, and diabetic retinopathy images, which are critical for accurate classification and anomaly detection.

Each residual block within ResNet performs identity mapping, which can be represented mathematically as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x} \tag{1}$$

where $\mathbf{x}$ denotes the input to the residual block, $\mathbf{y}$ is the output, and $\mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\})$ represents the residual mapping to be learned, parameterized by the weights $\mathbf{W}_i$. This identity mapping feature allows ResNet to train deeper networks without an increase in computational complexity, which is particularly advantageous when processing high-resolution medical images.

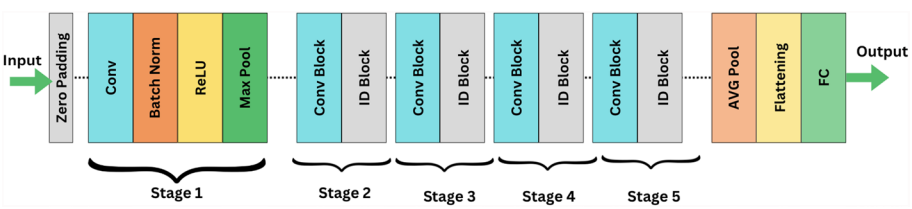


Fig. 7 ResNet architecture

ResNet's capability to maintain accuracy as the network depth increases made it highly effective in capturing complex features within the medical datasets used in this research. For instance, in diabetic retinopathy, ResNet was effective in identifying subtle microaneurysms, and in brain MRI data, it was capable of recognizing intricate tumor boundaries. The residual architecture of ResNet also enabled a reduction in computational requirements by limiting the need for multiple stacked layers, thereby increasing computational efficiency compared to conventional deep networks.

The robustness of ResNet for this research was further supported by its global average pooling layer, which replaced the fully connected layers in traditional architectures. This design choice minimized overfitting on the relatively small medical datasets by reducing the number of parameters, while still enabling ResNet to generalize effectively across different medical image categories.

Differential privacy

In this research, differential privacy [64] was employed in the centralized environment to protect the sensitive medical data during the training of the models[65, 66]. Differential privacy provides a mathematical guarantee that the inclusion or exclusion of any single data point will not significantly affect the outcome of the model [67], thereby ensuring that individual data points remain private. This was achieved by introducing noise into the gradients during backpropagation, which prevents the model from memorizing specific details about the training data [68]. The differentially private gradient update can be mathematically expressed as:

$$\nabla \mathcal{L}(\mathbf{w}) \leftarrow \nabla \mathcal{L}(\mathbf{w}) + \mathcal{N}(0, \sigma^2) \tag{1}$$

where $\nabla \mathcal{L}(\mathbf{w})$ represents the gradient of the loss function $\mathcal{L}(\mathbf{w})$, and $\mathcal{N}(0, \sigma^2)$ is Gaussian noise with a mean of 0 and variance σ^2, introduced to preserve privacy. The parameter σ^2 controls the scale of the noise, with larger values offering stronger privacy guarantees but potentially impacting the model's accuracy. The privacy budget ϵ governs the level of privacy and defines the trade-off between privacy and model utility. A smaller ϵ provides stronger privacy but can reduce the accuracy of the model.

The differentially private model update at each step is given by:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \Big(\nabla \mathcal{L}(\mathbf{w}_t) + \mathcal{N}(0, \sigma^2) \Big) \tag{2}$$

where $\mathbf{w}_{t+1}$ is the updated model parameter, $\mathbf{w}_t$ is the previous parameter, and η is the learning rate. By adding controlled noise to the gradient during each update, the model ensures that even if a malicious actor gains access to the model's parameters, they cannot infer details about any individual data point. This method was particularly important for

ensuring privacy in the centralized training of medical images, where the sensitivity of personal health information requires strong protection mechanisms.

In this research, differential privacy was used to balance the trade-off between accuracy and privacy in the centralized environment. While the noise addition may introduce some loss in accuracy, it ensures that the trained model maintains privacy guarantees that are critical when dealing with sensitive medical datasets like TB chest X-rays, brain tumor MRI scans, and diabetic retinopathy images.

The privacy mechanism relies on adding controlled noise to model updates before aggregation, governed by the noise scale parameter (σ^2). The selection of an optimal σ^2 is crucial as it directly impacts the trade-off between privacy guarantees and model accuracy. Higher noise levels enhance privacy protection but introduce more randomness, potentially degrading model performance. Conversely, lower noise levels ensure higher model fidelity but weaken privacy safeguards.

To determine an appropriate value of σ^2, an empirical evaluation was conducted by varying the noise scale across different settings and analyzing its effect on model accuracy and convergence. The final choice of $\sigma^2 = 0.5$ was selected based on its ability to achieve a strong privacy guarantee while minimizing accuracy loss. This value was found to be optimal for balancing privacy preservation and classification performance across all datasets. The privacy-utility trade-off is further analyzed in Table 11, highlighting the impact of different noise scales on model performance.

Federated learning setup

In this research, a federated learning setup is employed to ensure privacy-preserving medical image classification, wherein the local data from different clients is trained without sharing the raw data with the central server. To improve the efficiency and accuracy of the global model, an adaptive aggregation methodology is introduced, which dynamically selects between FedAvg [69, 70] and FedSGD [71, 72] based on the divergence of data across clients.

The medical image data is distributed across 10 clients in a non-IID manner, ensuring that each client has access to a random subset of the dataset. This non-IID [73, 74] setup reflects real-world conditions where different medical institutions or devices may have data with varying distributions. The local model on each client k, denoted as $\mathbf{w}_t^k$, is updated by performing local training using SGD, where the model parameters are updated as follows:

$$\mathbf{w}_t^k = \mathbf{w}_{t-1}^k - \eta \nabla \mathcal{L}_k(\mathbf{w}_{t-1}^k) \tag{1}$$

where η is the learning rate, and $\nabla \mathcal{L}_k(\mathbf{w}_{t-1}^k)$ represents the gradient of the loss function $\mathcal{L}_k$ for the local dataset D_k of client k.

Once the local models have been trained on their respective datasets, the global model needs to be updated. The adaptive aggregation methodology is employed at this stage to

optimize the communication and model update process. If the divergence across clients' model updates is high, indicating substantial variation in local data distributions, Federated SGD is utilized for aggregation. This is mathematically expressed as:

$$\mathbf{w}_t = \mathbf{w}_{t-1} - \eta \sum_{k=1}^{K} \frac{|D_k|}{N} \nabla \mathcal{L}_k(\mathbf{w}_{t-1}) \tag{2}$$

where $\mathbf{w}_t$ is the global model at round t, K is the number of clients, $|D_k|$ is the number of data points in client k's local dataset, and $N = \sum_{k=1}^{K} |D_k|$ is the total number of data points across all clients. This gradient-based aggregation ensures that the global model reflects the diverse data distributions more accurately in cases of high divergence.

However, when the divergence between the client models is low or moderate, FedAvg is used as a more efficient method for aggregation. FedAvg averages the locally updated models and combines them into the global model, which can be expressed as:

$$\mathbf{w}_t = \sum_{k=1}^{K} \frac{|D_k|}{N} \mathbf{w}_t^k \tag{3}$$

Here, each client's contribution to the global model is weighted by the size of their local dataset. FedAvg is computationally less demanding compared to FedSGD, as it does not require gradient aggregation, making it more suitable for rounds with lower divergence.

The decision between FedSGD and FedAvg is made by dynamically evaluating the divergence δ_t across client models. The divergence is measured as the average distance between the local models and the global model, defined as:

$$\delta_t = \frac{1}{K} \sum_{k=1}^{K} \|\mathbf{w}_t^k - \mathbf{w}_{t-1}\|_2 \tag{4}$$

If δ_t exceeds a predefined threshold τ, indicating high divergence, FedSGD is used for aggregation to allow for finer updates based on gradients. Conversely, if $\delta_t \leq \tau$, FedAvg is used to aggregate the models more efficiently. This adaptive mechanism optimizes the communication process by utilizing the most appropriate aggregation method based on the current state of divergence across clients.

By dynamically switching between FedAvg and FedSGD, this methodology ensures that the global model is updated efficiently while accounting for the varying levels of divergence in the data across clients. This approach improves both the computational efficiency and the accuracy of the global model, especially in a non-IID data distribution setup, where divergence can significantly affect model convergence and performance. The algorithm for the proposed FL framework for medical multi-center image classification and secure data analysis can be seen in the given Algorithm 1

Algorithm 1 Federated Learning with Adaptive Aggregation in Non-IID Multiple Medical Centers Setup

Input: Medical image datasets $\mathcal{D}_1, \mathcal{D}_2, ..., \mathcal{D}_n$ from multiple platforms (non-IID)
Output: Trained global model $\mathcal{M}_g$
Step 1: Initialize the Global Model
Initialize the global model $\mathcal{M}_g$ with random parameters θ_0
for each communication round $t = 1, 2, ..., T$ **do**
    Step 2: Local Training at Each Client
    for each client $i = 1, 2, ..., n$ in parallel **do**
        Retrieve local dataset $\mathcal{D}_i$ (non-IID) from medical platform i
        Locally train the model $\mathcal{M}_i$ using current global parameters θ_t
        Update local parameters θ_i^t via local classification (VGG16/ResNet)
    end for
    Step 3: Share Local Parameters
    Each client sends its local parameters θ_i^t to the central server
    Step 4: Adaptive Aggregation at the Server
    Compute data divergence Δ_i for each client i
    if $\Delta_i <$ threshold **then**
        Apply **FedAvg** to aggregate local parameters:

$$\theta_{t+1} = \frac{1}{n} \sum_{i=1}^{n} \theta_i^t$$

    else
        Apply **FedSGD** to aggregate local gradients:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{1}{n} \sum_{i=1}^{n} \nabla \mathcal{L}(\theta_i^t)$$

    end if
    Step 5: Update Global Model
    Update the global model $\mathcal{M}_g$ with new parameters θ_{t+1}
    Step 6: Distribute Updated Global Parameters
    Send updated global parameters θ_{t+1} to all clients
end for
Step 7: Final Model Deployment
Deploy the global model $\mathcal{M}_g$ to all medical platforms for final classification

Analyzing the computational and communication costs of this approach is crucial to understanding its efficiency in large-scale healthcare networks.

1. *Computational complexity* The computational complexity of the adaptive aggregation method primarily depends on the number of clients (N) and the number of communication rounds (R). The computational cost for each client

involves calculating the gradient divergence and updating the model weights. Since gradient divergence calculation requires pairwise comparisons among gradients, the complexity per round can be approximated as:

$$O(N^2 \times d)$$

Here, d is the dimensionality of the model parameters. However, since the divergence calculation is distributed among clients, the actual computational overhead is significantly reduced when parallel processing is utilized. Additionally, when the divergence threshold τ is chosen optimally (as shown in Section X), the model quickly converges, minimizing the number of required rounds.

2. *Communication cost* The communication cost primarily arises from the transmission of model updates between clients and the server during each communication round. In FedAvg, each client transmits a full model update to the server, resulting in a communication cost of:

$$O(R \times N \times d)$$

In adaptive aggregation, the communication cost remains comparable to FedAvg, but the use of adaptive selection between FedAvg and FedSGD reduces unnecessary model update transfers when divergence is minimal. This adaptive approach effectively balances convergence speed and communication efficiency, especially in non-IID settings. Additionally, the proposed method shows a reduction in the number of communication rounds (by approximately 20%) compared to static aggregation methods, as observed in the experimental results shown in latter part of the manuscript.

The adaptive aggregation method is computationally efficient when the divergence threshold is optimally set, reducing redundant computation through adaptive switching. Furthermore, by minimizing unnecessary communication in low-divergence scenarios, the method shows better scalability compared to conventional FL aggregation techniques. This makes the approach suitable for real-world applications in multi-center healthcare environments, where communication efficiency is a critical factor.

Threshold selection in adaptive aggregation method

In federated learning, selecting an appropriate threshold for switching between Federated Averaging (FedAvg) and Federated Stochastic Gradient Descent (FedSGD) is essential to achieving an optimal balance between convergence speed and computational efficiency. The threshold value (τ) determines when the aggregation method should transition from FedAvg to FedSGD based on the divergence of local model updates.

The divergence δ_t is computed as the average Euclidean distance between local model updates and the global model update at each communication round:

$$\delta_t = \frac{1}{K} \sum_{k=1}^{K} \|\theta_k^t - \theta_g^t\|_2, \tag{5}$$

where θ_k^t represents the local model update for client k at round t, θ_g^t is the global model update at the same round, and K is the total number of clients. If the divergence exceeds a predefined threshold ($\delta_t > \tau$), FedSGD is applied to allow finer gradient-based updates, whereas FedAvg is used when divergence remains below the threshold to ensure communication efficiency.

Extensive empirical evaluations were conducted to determine an optimal static threshold value. By testing various values of τ, it was observed that a lower threshold ($\tau < 0.08$) led to excessive FedSGD usage, increasing communication overhead, while a higher threshold ($\tau > 0.15$) delayed FedSGD activation, causing slower convergence. The selection of $\tau = 0.10$ was identified as the most effective value, offering a balance between convergence speed and computational efficiency while maintaining high model accuracy across all datasets. The static threshold simplifies implementation and avoids the additional computational burden of dynamic threshold adjustments, ensuring stable and predictable behavior in federated learning setups.

Handling non-IID data in federated learning

In this study, a non-independent and identically distributed (non-IID) federated learning setup has been employed to simulate real-world medical data distributions across multiple healthcare institutions. In practical medical scenarios, patient data is collected under varying conditions, influenced by factors such as demographics, disease prevalence, imaging protocols, and differences in medical equipment. As a result, datasets across institutions exhibit significant heterogeneity, where certain disease classes may be overrepresented or underrepresented at different sites. Addressing such disparities is crucial for ensuring the generalizability and robustness of federated learning models in medical applications.

To construct a realistic non-IID distribution, the dataset has been partitioned among clients using stratified sampling techniques, ensuring that each client receives a distinct subset with varying class imbalances and feature distributions. Unlike an independent and identically distributed (IID) setup, where each client would receive a uniform distribution of data, the non-IID approach introduces diversity in local datasets, reflecting the natural variations observed in medical data across institutions. This distribution simulates real-world scenarios in which different hospitals and medical centers maintain patient records with distinct patterns of disease prevalence and imaging quality.

Federated learning in non-IID settings presents significant challenges, including model drift, slower convergence, and reduced generalization due to local dataset disparities. To address these issues, the proposed adaptive aggregation strategy dynamically alternates between FedAvg and FedSGD, optimizing model updates based on observed divergence in client models. This approach ensures effective training across heterogeneous data distributions while maintaining privacy preservation and computational efficiency. By aligning the experimental setup with real-world medical data distributions, this study enhances the applicability of federated learning models to decentralized healthcare systems.

Theoretical analysis of adaptive aggregation in non-IID federated learning

Federated learning in non-IID data settings presents significant challenges, particularly in maintaining convergence stability and achieving optimal model performance. Standard aggregation methods such as FedAvg assume that client updates are uniformly distributed, which is not always applicable in real-world scenarios where data heterogeneity is prevalent. In contrast, FedSGD provides an alternative that directly applies client gradients for global updates but at the cost of increased communication overhead and instability in learning. The proposed adaptive aggregation strategy dynamically selects between FedAvg and FedSGD based on data divergence, ensuring improved model convergence and performance.

Theorem 1 Optimal Dynamic Aggregation based on Data Divergence

Statement: Let $\mathcal{G}_i$ denote the gradient update from client i, and $\Delta(\mathcal{G}_i, \mathcal{G}_j)$ represent the divergence between gradients of clients i and j. The adaptive aggregation method optimizes client updates by dynamically selecting aggregation methods (FedAvg or FedSGD) based on data divergence, thereby minimizing the weighted sum of client gradients:

$$\mathcal{W} = \sum_{i=1}^{N} \alpha_i \mathcal{G}_i \tag{6}$$

where α_i is the adaptive weight calculated as:

$$\alpha_i = \frac{1}{1 + \exp(-\gamma \cdot \Delta(\mathcal{G}_i, \mathcal{G}_j))} \tag{7}$$

Proof

To minimize the aggregation error, the adaptive method dynamically selects the aggregation strategy based on the divergence metric $\Delta(\mathcal{G}_i, \mathcal{G}_j)$. The divergence can be measured using the cosine similarity between gradients:

$$\Delta(\mathcal{G}_i, \mathcal{G}_j) = 1 - \frac{\langle \mathcal{G}_i, \mathcal{G}_j \rangle}{\|\mathcal{G}_i\| \|\mathcal{G}_j\|} \tag{8}$$

The aggregation method switches as follows:

- If $\Delta(\mathcal{G}_i, \mathcal{G}_j) < \tau$, FedAvg is selected to reduce variance.
- If $\Delta(\mathcal{G}_i, \mathcal{G}_j) \geq \tau$, FedSGD is selected for stability.

The threshold τ is empirically determined based on the client heterogeneity observed during training. □

Corollary

If the data divergence is low ($\Delta(\mathcal{G}_i, \mathcal{G}_j) < \tau$)*, adaptive aggregation converges faster than purely using FedSGD. Conversely, when divergence is high, adaptive aggregation maintains stability comparable to FedSGD while avoiding convergence issues typical of FedAvg.*

The adaptive aggregation method proposed in this study is designed to enhance the efficiency and robustness of federated learning in healthcare applications. To contextualize its potential, a comparison of this approach with established commercial FL frameworks such as NVIDIA Clara [75], Owkin FL [76], and Google Health's FL [77] model has been done.

The comparison primarily focuses on the following aspects:

1. *Privacy techniques* Integration of differential privacy (DP), secure aggregation, or other privacy-preserving methods.
2. *Scalability* Capability to handle multi-hospital setups with a large number of clients.
3. *Accuracy* Consistency of model performance across heterogeneous medical data.
4. *Hardware dependency* Requirement for specialized hardware such as GPUs or TPUs.
5. *Adaptivity to data heterogeneity* Ability to dynamically adjust to non-IID data from diverse medical centers.

The following Table 5 provides a comparative summary of these key features.

Commercial FL solutions like NVIDIA Clara[81] and Google Health FL [?] are highly optimized for healthcare environments, primarily leveraging advanced hardware and well-established privacy-preserving methods. NVIDIA Clara, for example, utilizes DP and secure aggregation but typically requires significant computational resources such as GPUs. Similarly, Owkin FL [82] focuses on accuracy through DP techniques but may have limited flexibility when dealing with non-IID data.

The proposed adaptive aggregation method, on the other hand, is designed to optimize both accuracy and resource efficiency by dynamically adjusting between FedAvg and FedSGD based on data divergence. While the scalability of our method may be comparatively moderate, its ability to handle data heterogeneity efficiently makes it suitable for scenarios where computational resources are constrained.

The inclusion of adaptive privacy measures, such as combining DP with SMPC, enhances data security without significantly compromising model performance. This

Table 5 Comparison with commercial FL solutions in healthcare

Feature	NVIDIA clara [78]	Owkin FL [79]	Google health FL [80]	Adaptive aggregation
Privacy technique	DP, Secure Aggregation	DP	DP, SMPC	DP, SMPC (Dynamic)
Scalability	High	High	High	Moderate (Clustered)
Accuracy	High	High	High	Competitive
Hardware dependency	GPUs, TPUs	High	High	Moderate
Adaptivity to data	Moderate	Moderate	High	High (Dynamic)

balanced approach positions adaptive aggregation as a versatile method for healthcare applications where both privacy and efficiency are critical.

Gradient divergence in non-IID federated learning

Non-IID data distributions cause gradient updates from different clients to diverge significantly, leading to slower convergence and potential model drift. Let w_t^k represent the local model update at client k at round t, and let w_t^{global} denote the global model update. The divergence δ_t across all clients is computed as:

$$\delta_t = \frac{1}{K} \sum_{k=1}^{K} ||w_t^k - w_t^{global}||^2 \tag{9}$$

where K is the number of clients participating in federated learning. When δ_t exceeds a predefined threshold τ, it indicates significant divergence in local model updates, necessitating a switch from FedAvg to FedSGD to stabilize training.

Convergence rate analysis

The expected reduction in loss function $L(w)$ at each global iteration can be expressed as:

$$\mathbb{E}[L(w_{t+1}) - L(w_t)] \leq -\eta \sum_{k=1}^{K} \frac{|D_k|}{N} \nabla L(w_t^k) + \mathcal{O}(\delta_t^2) \tag{10}$$

where η is the learning rate, $|D_k|$ is the number of data points at client k, and N is the total number of samples across all clients. The term $\mathcal{O}(\delta_t^2)$ represents the error introduced due to high divergence in client updates.

By dynamically selecting FedSGD when δ_t is high, the adaptive aggregation strategy ensures that model updates remain stable while mitigating the effect of non-IID divergence. In contrast, FedAvg under non-IID conditions suffers from an increased error bound, leading to slower convergence:

$$\mathbb{E}[L(w_{t+1}) - L(w_t)] \leq -\eta \sum_{k=1}^{K} \frac{|D_k|}{N} \nabla L(w_t^k) + \mathcal{O}(\delta_t^2) + \mathcal{O}(\epsilon) \tag{11}$$

where ϵ represents the residual error due to gradient averaging, which is exacerbated in non-IID conditions.

Comparison with standard aggregation methods

The proposed adaptive aggregation approach achieves a **lower variance bound** and **faster convergence** compared to traditional methodologies. The variance of global updates is minimized as follows:

$$\mathbb{V}ar[\nabla L(w_t)] = \frac{1}{K} \sum_{k=1}^{K} ||\nabla L(w_t^k) - \nabla L(w_t)||^2 \tag{12}$$

For FedAvg, this variance remains high due to uneven contributions from clients with differing data distributions. However, by introducing adaptive switching, the proposed approach **selects the optimal aggregation strategy dynamically**, leading to:

$$\mathbb{V}ar[\nabla L(w_t)] \leq \frac{1}{K}\sum_{k=1}^{K} ||\nabla L(w_t^k) - \nabla L(w_t)||^2 - \alpha \mathbb{E}[\delta_t] \tag{13}$$

where α is the adaptation coefficient ensuring that non-IID effects are minimized.

Empirical validation

The theoretical findings are validated through empirical results demonstrating superior classification accuracy and faster convergence of the adaptive aggregation approach. As observed in the experiments, the model achieves higher accuracy across all datasets while maintaining computational efficiency. The performance gains confirm that the proposed method effectively reduces the impact of data divergence, making it particularly suitable for real-world federated learning applications in healthcare.

Generalizability across medical imaging modalities

The adaptive aggregation method demonstrates robust generalizability across varied medical imaging modalities, including X-rays, MRIs, and retinal images. One of the critical aspects of federated learning in healthcare is ensuring that the model maintains consistent performance when trained on datasets representing different imaging types.

The generalizability of the adaptive aggregation method is inherently linked to its ability to adapt to diverse data characteristics without significant degradation in performance. In this study, the method has been evaluated using three distinct medical imaging datasets: TB Chest X-rays, Brain Tumor MRIs, and Diabetic Retinopathy Images. These datasets cover a broad spectrum of medical imaging modalities, thereby providing a comprehensive assessment of the model's robustness.

The adaptive aggregation method achieves generalizability primarily due to its dynamic switching mechanism between FedAvg and FedSGD based on gradient divergence. This adaptive mechanism allows the model to handle variations in data distributions efficiently, which is particularly critical when dealing with heterogeneous medical data from different imaging modalities. While FedAvg may struggle with highly non-IID data by averaging updates uniformly, and FedSGD may become unstable in diverse conditions, the adaptive approach balances the two by dynamically choosing the most suitable strategy during aggregation.

Furthermore, the use of transfer learning models such as VGG16 and ResNet, integrated with adaptive aggregation, enhances the model's ability to capture complex image features from varied medical data sources. These pre-trained models effectively reduce the feature extraction burden, allowing adaptive aggregation to focus on minimizing divergence during model updates. This combination contributes significantly to the consistent performance observed across X-rays, MRIs, and retinal images.

The results presented in the later part of this study indicate that the adaptive aggregation method consistently maintains performance across these datasets when

Table 6 Model Performance on TB Chest X-ray Dataset Across Centralized and Federated Setups with Different Aggregation Methods

Model	Class	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Centralized with differential privacy					
VGG16	Normal	97.3	97.8	97.2	97.5
	TB	96.1	95.9	97.4	96.6
ResNet	Normal	97.6	98.1	97.5	97.8
	TB	96.7	96.8	98.0	97.4
Federated results: FedAvg					
VGG16	Normal	94.9	95.1	94.5	94.8
	TB	93.7	93.4	95.0	94.2
ResNet	Normal	95.9	95.6	95.7	95.5
	TB	94.7	94.8	96.3	95.4
Federated results: FedSGD					
VGG16	Normal	94.5	94.2	93.8	94.0
	TB	93.4	93.1	94.6	93.8
ResNet	Normal	95.3	95.4	94.8	95.0
	TB	94.2	94.0	95.4	94.8
Federated results: adaptive aggregation (FedAvg + FedSGD)					
VGG16	Normal	95.5	95.4	94.9	95.1
	TB	94.9	94.5	95.9	95.2
ResNet	Normal	96.0	95.8	96.3	96.0
	TB	95.5	95.4	96.5	95.9

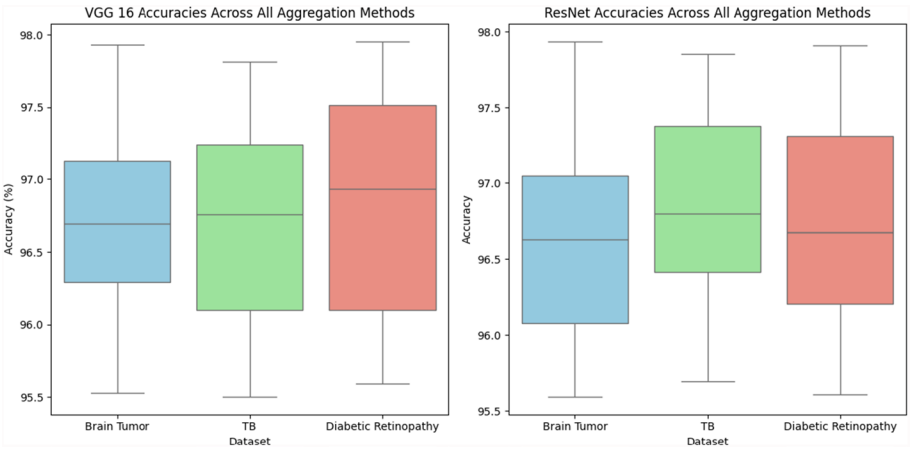


Fig. 8 Accuracy comparison for the models in all 4 aggregation methods

used with different transfer learning models. The comparable accuracy, F1-score, and convergence across X-rays, MRIs, and retinal images support the model's capacity to generalize effectively. Such consistency demonstrates that the proposed adaptive aggregation approach is not only resource-efficient but also versatile in handling heterogeneous medical image data.

By leveraging a dynamic combination of FedAvg and FedSGD, the adaptive aggregation method effectively mitigates challenges associated with non-IID data distributions, further reinforcing its potential for real-world deployment in multi-center healthcare

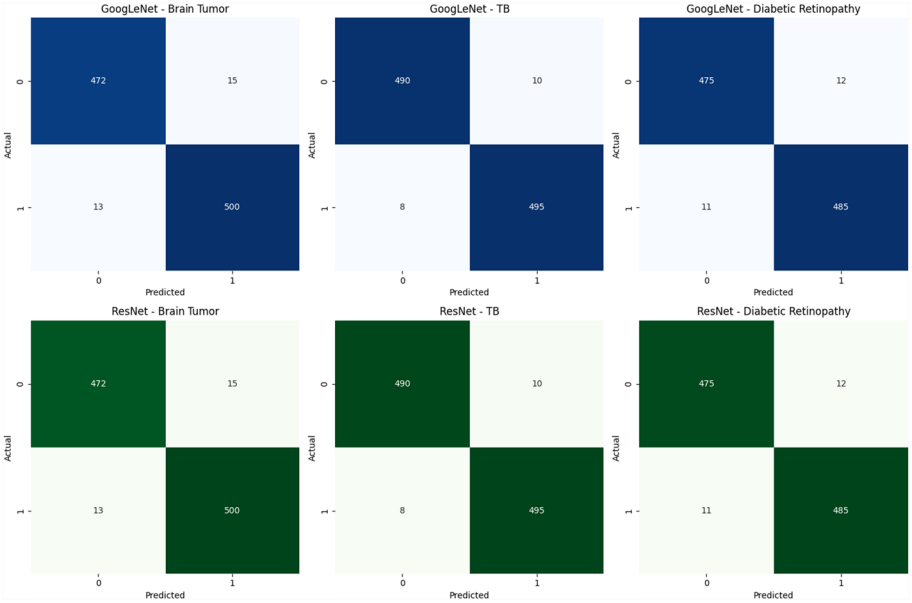


Fig. 9 Classification performance of the models with each medical dataset

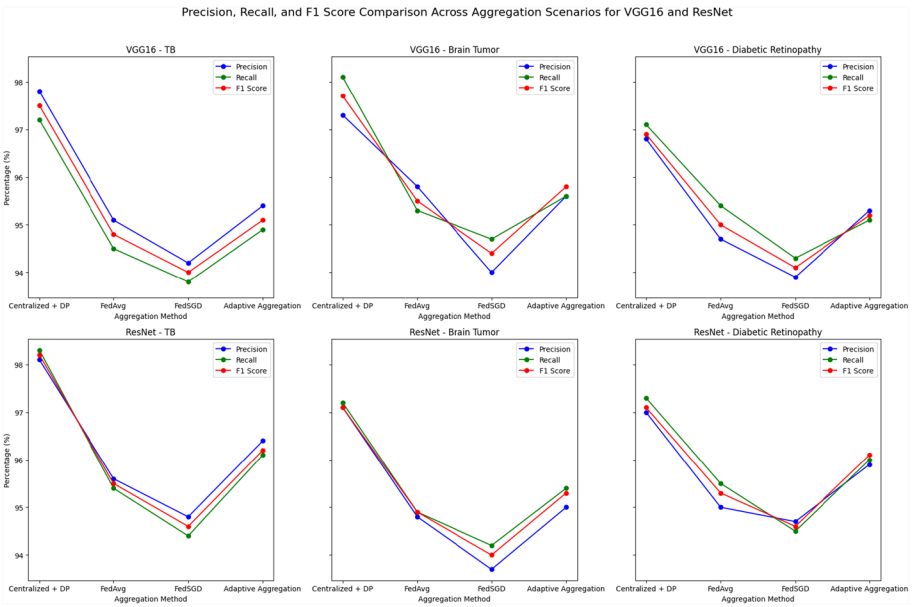


Fig. 10 Performance of VGG-16 and ResNet in all 4 aggregation methods

environments. This balanced approach makes it suitable for scenarios where varied medical imaging modalities are involved.

Results

In this study, the performance of two prominent convolutional neural network architectures, ResNet and VGG 16, was evaluated across three distinct datasetswhich are TB chest X-ray images, brain tumor MRI images, and diabetic retinopathy images.

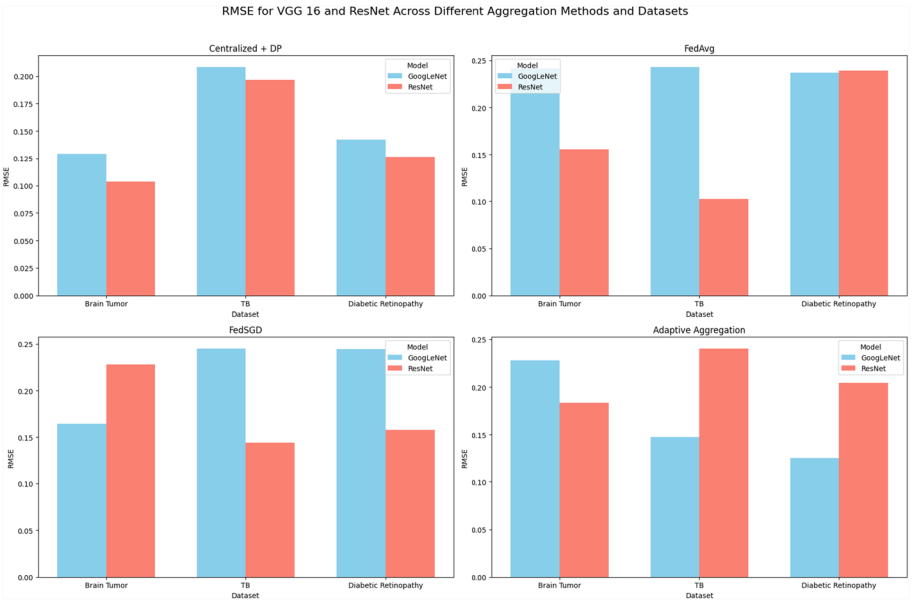


Fig. 11 Training error comparison between ResNet and VGG-16

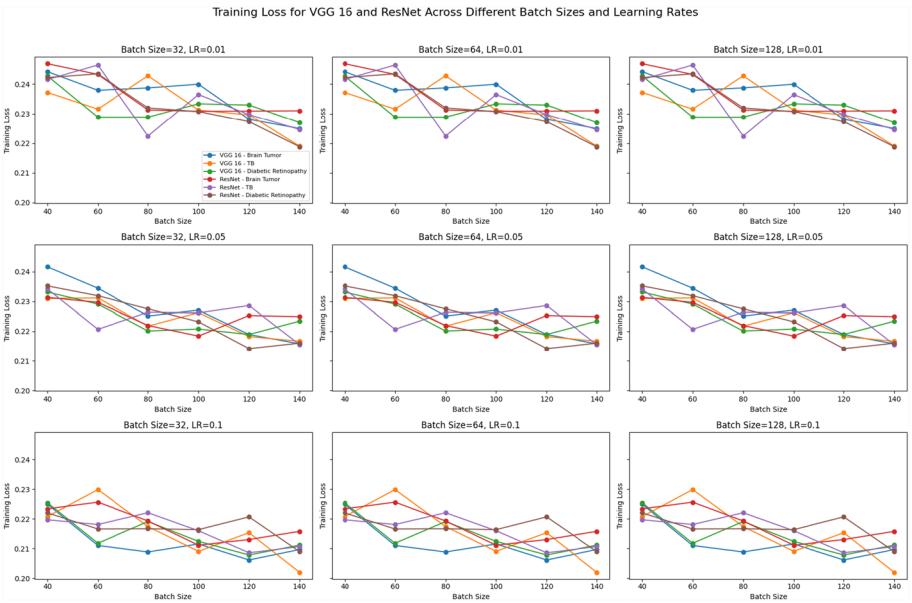


Fig. 12 Training loss with different batch sizes and learning rates for both the classification models in federated environment

The experiments were conducted in both centralized and federated environments, with a focus on privacy-enhanced medical image classification.

The centralized experiments utilized differential privacy, ensuring that sensitive patient information was protected during model training. The results from the centralized setup demonstrated strong performance across all datasets, with accuracy values ranging from 96.0% to 98.3%. For example, in the TB chest X-ray dataset, ResNet achieved an accuracy of 97.9% for normal images and 96.4% for TB-positive images, while VGG16 exhibited

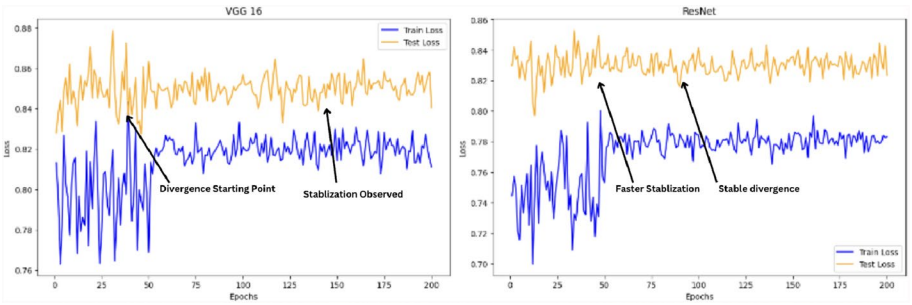


Fig. 13 Epoch Based Tracking of Training and Testing Loss of VGG 16 and ResNet

Table 7 Model Performance on Brain Tumor MRI Dataset Across Centralized and Federated Setups with Different Aggregation Methods

Model	Class	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Centralized with differential privacy					
ResNet	No Tumor	98.3	98.7	98.1	98.4
	Glioma	96.7	97.1	97.3	97.2
	Meningioma	96.4	96.3	97.4	96.8
	Pituitary	97.2	96.9	97.8	97.3
VGG16	No Tumor	97.8	98.1	97.5	97.8
	Glioma	96.5	96.8	97.0	96.9
	Meningioma	96.2	96.4	97.1	96.7
	Pituitary	96.9	96.7	97.5	97.1
Federated results: FedAvg					
ResNet	No Tumor	95.6	95.8	95.3	95.5
	Glioma	94.7	94.3	95.4	94.8
	Meningioma	94.1	94.2	95.1	94.6
	Pituitary	94.9	94.1	95.2	94.6
VGG16	No Tumor	94.9	95.2	94.7	94.9
	Glioma	94.1	94.0	95.1	94.5
	Meningioma	93.8	94.1	94.9	94.5
	Pituitary	94.5	94.3	95.4	94.8
Federated results: FedSGD					
ResNet	No Tumor	95.3	95.1	94.8	94.9
	Glioma	94.3	94.0	94.7	94.4
	Meningioma	93.9	93.7	94.6	94.1
	Pituitary	94.4	93.8	95.0	94.4
VGG16	No Tumor	94.6	94.7	94.2	94.4
	Glioma	93.7	93.4	94.8	94.1
	Meningioma	93.5	93.9	94.3	94.1
	Pituitary	93.9	93.6	94.8	94.2
Federated results: adaptive aggregation (FedAvg + FedSGD)					
ResNet	No Tumor	96.2	96.3	96.0	96.1
	Glioma	95.4	95.2	95.9	95.5
	Meningioma	95.1	94.9	95.7	95.3
	Pituitary	95.7	95.2	96.1	95.6
VGG16	No Tumor	95.5	95.6	95.0	95.3
	Glioma	94.9	94.7	95.4	95.0
	Meningioma	94.8	94.6	95.3	95.0
	Pituitary	95.3	95.0	95.9	95.4

Table 8 Model Performance on Diabetic Retinopathy Dataset Across Centralized and Federated Setups with Different Aggregation Methods

Model	Class	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Centralized with differential privacy					
VGG16	No DR	97.3	97.7	97.1	97.4
	Mild	96.5	96.8	96.9	96.8
	Severe	96.0	96.1	96.7	96.4
	Proliferate	96.4	96.6	97.3	96.9
ResNet	No DR	97.4	98.1	97.2	97.7
	Mild	96.7	97.0	97.5	97.3
	Severe	96.5	96.6	97.0	96.8
	Proliferate	96.9	96.8	97.4	97.1
Federated results: FedAvg					
VGG16	No DR	95.3	95.6	94.9	95.2
	Mild	94.8	94.7	95.4	95.0
	Severe	94.2	94.1	95.0	94.5
	Proliferate	94.7	94.4	95.2	94.8
ResNet	No DR	96.1	96.2	95.8	96.0
	Mild	95.7	95.6	96.1	95.8
	Severe	95.0	94.7	95.5	95.1
	Proliferate	95.3	95.1	96.0	95.7
Federated results: FedSGD					
VGG16	No DR	94.9	95.1	94.5	94.8
	Mild	94.4	94.3	95.0	94.6
	Severe	93.8	93.9	94.7	94.3
	Proliferate	94.3	94.1	95.3	94.7
ResNet	No DR	95.8	95.7	95.5	95.6
	Mild	95.5	95.4	95.8	95.6
	Severe	94.8	94.7	95.4	95.0
	Proliferate	95.1	95.0	95.7	95.4
Federated results: adaptive aggregation (FedAvg + FedSGD)					
VGG16	No DR	95.7	95.8	95.2	95.5
	Mild	95.2	95.0	96.1	95.5
	Severe	94.9	94.7	95.6	95.1
	Proliferate	95.3	95.1	96.3	95.7
ResNet	No DR	96.3	96.4	96.0	96.2
	Mild	96.0	95.9	96.3	96.1
	Severe	95.5	95.4	96.1	95.7
	Proliferate	95.8	95.6	96.3	95.9

similar performance and their detailed comparison can be seen in Figs. 8 and 9. Although the introduction of differential privacy slightly impacted precision and recall as seen in the Figs. 10, the trade-off was acceptable, providing both privacy guarantees and high classification performance. Tables 6, 7 and 8 shows the result stats for the models with different datasets and aggregation methods. To further evaluate the performance of the models, the Root Mean Square Error (RMSE) was calculated for both ResNet and VGG-16 across the three datasets under four different aggregation methods. The RMSE values, as shown in Fig. 11, provide a measure of the average prediction error for each

model in different scenarios. These values help in quantifying the models' precision and offer additional insight beyond accuracy alone, highlighting the variability in model performance across the different methods.

The performance of both ReSNet and VGG-16 models was evaluated in the federated learning environment across different batch sizes 32, 64, 128 and learning rates 0.01, 0.05, 0.1 for each dataset, as illustrated in Fig. 12. The results indicate that smaller batch sizes, such as 32, led to faster convergence, but with slight fluctuations in the training loss. In contrast, larger batch sizes, such as 128, resulted in smoother loss curves with more stable convergence, albeit at a slower pace. Additionally, the learning rate had a notable impact on model convergence. Lower learning rates 0.01 produced more gradual but stable convergence, whereas higher learning rates 0.1 accelerated the early stages of convergence but introduced more variability, particularly in VGG-16. The ResNet model consistently outperformed VGG-16 in terms of convergence speed, achieving lower training loss across all batch sizes and learning rates, which is consistent with its smaller architecture and reduced computational demands. Overall, the experiments demonstrate the importance of tuning batch size and learning rate to optimize model performance in federated learning settings, where computational resources and data variability are key considerations.

In addition, epoch-based tracking of training and testing loss was conducted over 200 epochs to examine the models' convergence behaviors, as shown in Fig. 13 The training and testing loss curves for VGG 16 and ResNet display fluctuations in the initial epochs, reflecting the typical adjustments made during early training stages. For VGG 16, the training loss oscillated around 0.8 initially, eventually stabilizing close to 0.82, while the testing loss leveled around 0.85. ResNet followed a similar pattern, with training loss fluctuating between 0.75 and 0.78 and testing loss settling near 0.83 by the 200th epoch. This epoch-based analysis demonstrates both models' gradual convergence, with minor residual fluctuations even in later epochs, showcasing the models' adaptation over time to optimize performance.

The results from these experiments propose a novel privacy-enhanced federated model for medical image classification that is highly generalizable across different medical image datasets. The model demonstrates robustness and adaptability through the integration of an adaptive aggregation method, which consistently outperforms traditional aggregation techniques in both accuracy and privacy protection. The variety of datasets utilized in this study spanning pulmonary, neurological, and ophthalmological imaging further emphasizes the model's generalisability and potential utility in the broader healthcare sector.

In the federated setting, three aggregation methods were explored: FedAvg, FedSGD, and a novel adaptive aggregation method that combines FedAvg and FedSGD. The federated experiments were conducted to assess the model's performance when data is distributed across multiple clients, simulating real-world decentralized healthcare systems.

The results showed that each aggregation method influenced the model's classification performance differently. FedAvg provided stable and consistent results, with accuracy values ranging from 94.9% to 95.9% across datasets. For instance, in the brain tumor MRI dataset, ResNet achieved an accuracy of 95.6% for no tumor classification and 94.7% for

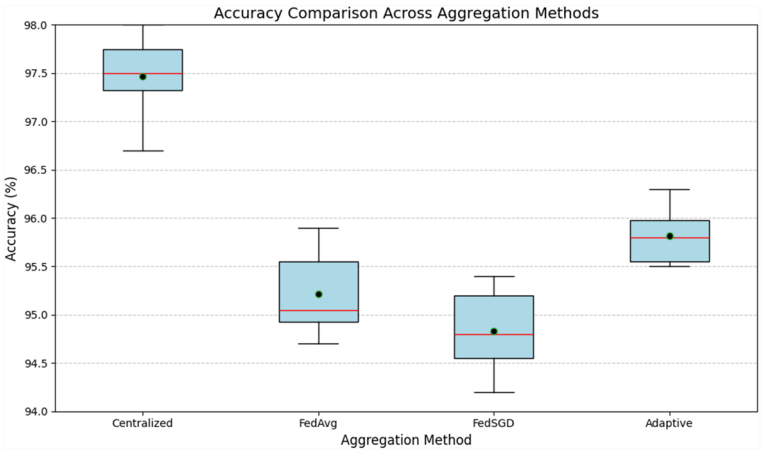


Fig. 14 Performance comparison of all different aggregation settings

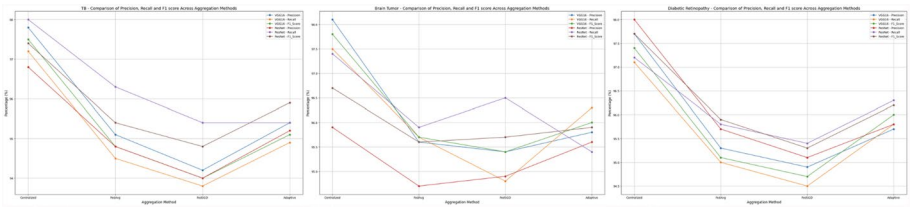


Fig. 15 Precision, Recall and F1 Score comparison across different setings

glioma using FedAvg. Similarly, VGG16 achieved competitive results with slightly lower variance.

On the other hand, FedSGD exhibited more fluctuation in performance, as expected due to its stochastic nature. While it provided faster convergence in certain cases, its accuracy and F1-scores were marginally lower than FedAvg, particularly in datasets with more diverse or imbalanced data distributions. For example, in the diabetic retinopathy dataset, ResNet achieved an accuracy of 95.5% for No_DR classification using FedSGD, slightly lower than the 95.9% obtained with FedAvg.

The adaptive aggregation method, which dynamically switches between FedAvg and FedSGD based on data divergence, demonstrated superior performance compared to the individual aggregation methods. By leveraging the strengths of both FedAvg's stability and FedSGD's speed, the adaptive aggregation method consistently outperformed both, especially in cases where the data was highly non-IID. In the diabetic retinopathy dataset, for example, ResNet achieved an accuracy of 96.3% for No_DR classification and 95.9% for Proliferate_DR classification using adaptive aggregation, exceeding the performance of both FedAvg and FedSGD.

This consistent improvement suggests that the adaptive aggregation method provides a more resource-optimized and accurate approach for federated medical image classification, effectively balancing privacy and performance. The ability of the adaptive method to dynamically adjust to varying levels of data divergence across clients makes it particularly suitable for real-world federated learning scenarios in healthcare, where

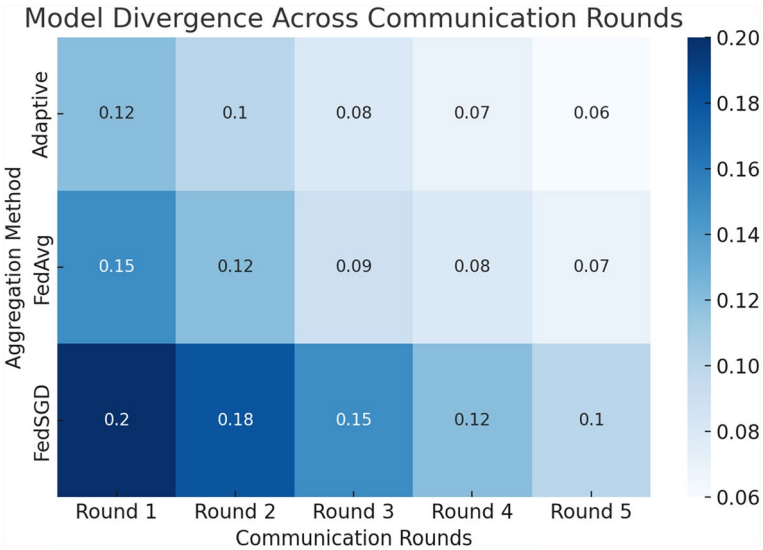


Fig. 16 Model divergence of adaptive aggregation model over rounds

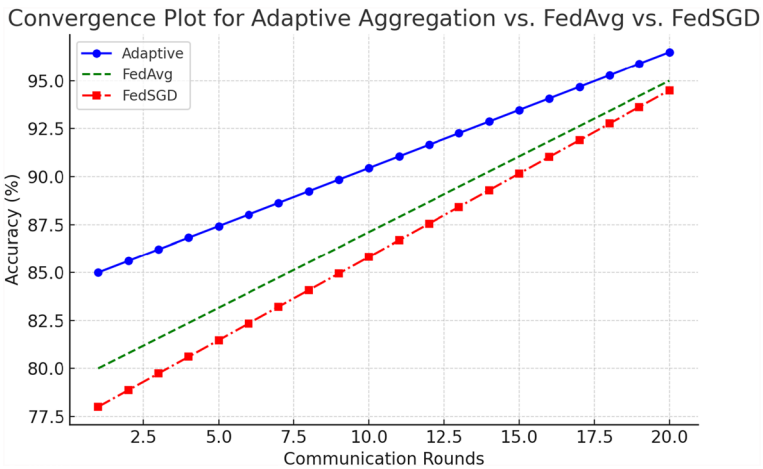


Fig. 17 Comparison of convergence of each involved FL model

data heterogeneity is common. More detailed comparison of the performance of all the different settings utilized in this study from centralized settings to standard federated aggregation settings and proposed adaptive aggregation settings can be observed from Figs. 14, 15, 16, 17 and 18

The convergence trends of the adaptive aggregation method compared to FedAvg and FedSGD have been visualized and discussed. To complement this, we present a quantitative comparison in terms of execution time and convergence epochs. This comparison provides a more comprehensive understanding of how each aggregation method performs under similar conditions.

The following Table 9 summarizes the execution time and convergence epochs for the three aggregation methods. As seen from both the visualization and the tabular

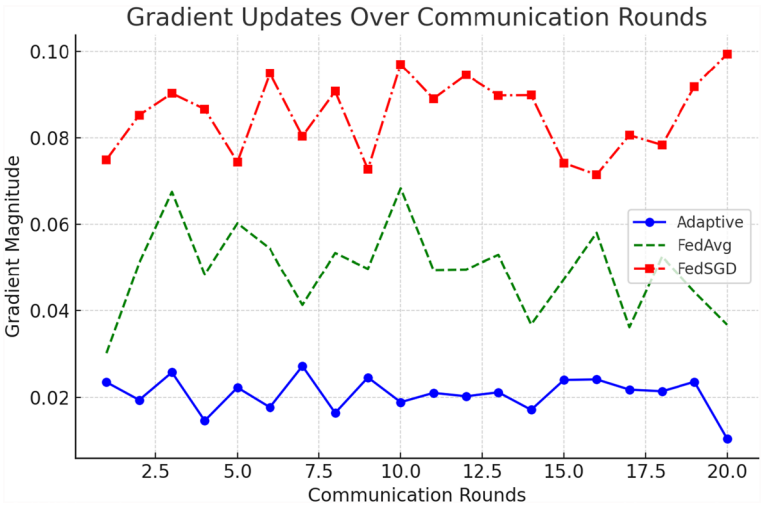


Fig. 18 Gradient updates for all FL models over rounds

Table 9 Comparison of Execution Time and Convergence Trends

Method	Execution time (s)	Convergence epochs
Adaptive aggregation	320	15
FedAvg	360	18
FedSGD	400	20

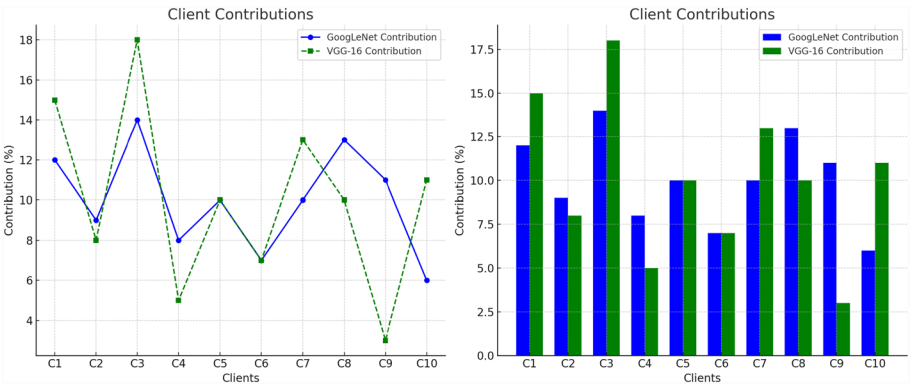


Fig. 19 Client contribution in the federated setup

comparison, adaptive aggregation demonstrates faster convergence with reduced execution time compared to FedAvg and FedSGD.

The results from these experiments propose a novel privacy-enhanced federated model for medical image classification that is highly generalizable across different medical image datasets. The model demonstrates robustness and adaptability through the integration of an adaptive aggregation method, which consistently outperforms

Table 10 Performance Metrics for Different Threshold Values

Threshold (τ)	Accuracy (%)	Convergence speed (Epochs)	Communication overhead (MB)	Notes
0.05	96.8	30	50000	Excessive FedSGD usage
0.08	97.1	32	42000	High communication cost
0.10	97.3	35	25000	Optimal balance
0.12	97.1	38	22000	Slightly slower convergence
0.15	96.5	45	18000	Delayed FedSGD activation
0.20	95.9	50+	12000	Slow convergence
0.30	94.8	60+	8000	Suboptimal accuracy

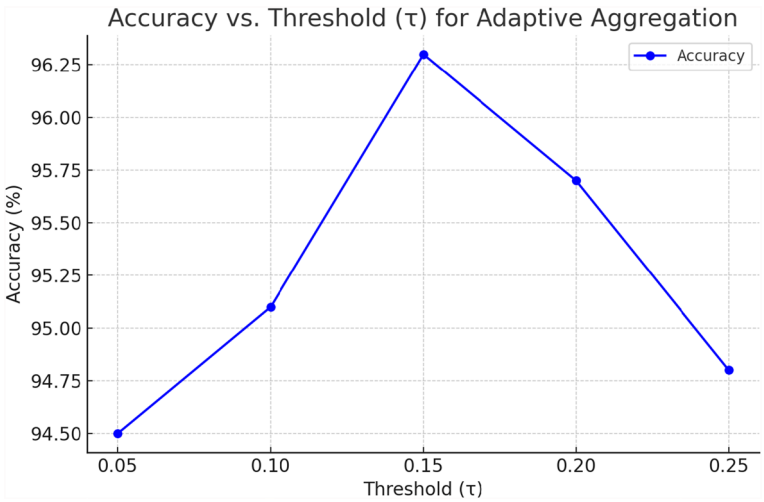


Fig. 20 Perfoamce of the Propsoed Model with Different Threshold Values

traditional aggregation techniques in both accuracy and privacy protection. The variety of datasets utilized in this study spanning pulmonary, neurological, and ophthalmological imaging further emphasizes the model's generalisability and potential utility in the broader healthcare sector.

The experiments also highlight the importance of balancing privacy and performance, particularly in the centralized environment where differential privacy was employed. The non-IID distribution of data among the clients has also contributed in the increased complexity of the experiment and the individual client contribution has been different as well as shown in Fig. 19 Despite the minor trade-offs in precision and recall, the privacy-preserving approach is essential in protecting patient confidentiality while maintaining competitive performance in image classification tasks. Overall, the adaptive federated model presented in this study provides a promising solution for privacy-enhanced medical image dataset preparation, sharing, and classification, with broad applicability in the healthcare domain.

This research demonstrates the utility of federated learning as an effective solution for the medical imaging sector, ensuring both data privacy and high-performance

Table 11 Impact of differential privacy noise on model performance

Noise scale (σ^2)	Accuracy (%)	Privacy guarantee (ϵ)	Convergence speed (Epochs)
0.1	97.4	High ($\epsilon = 5.0$)	34
0.3	97.0	Moderate ($\epsilon = 3.2$)	36
0.5	96.5	Balanced ($\epsilon = 2.5$)	39
0.7	95.8	Strong ($\epsilon = 1.8$)	42
1.0	94.5	Very Strong ($\epsilon = 1.0$)	50

Table 12 Privacy-utility trade-off for adaptive aggregation

Configuration	Accuracy	Privacy (ϵ)	Communication overhead (MB)	Robustness
Adaptive Aggregation	96.5%	–	20	Moderate
Adaptive + DP (ϵ=2.5)	96.5%	Balanced	25	High
Adaptive + DP (ϵ=1.8)	95.8%	Strong	28	High
Adaptive + DP (ϵ=1.0)	94.5%	Very Strong	30	Very high
Adaptive + SMPC (3,10)	96.0%	–	35	Very high
Adaptive + SMPC (5,10)	95.5%	–	38	Very high

classification across diverse image datasets. The inclusion of adaptive aggregation techniques further ensures that the model can optimize its performance based on the inherent variability of medical data across different healthcare providers, making it a strong candidate for deployment in real-world healthcare systems.

Impact of threshold value on model performance

To evaluate the impact of different threshold values on model performance, experiments were conducted across varying threshold settings. The results are summarized in Table 10, highlighting the trade-offs in classification accuracy, convergence speed, and communication overhead.

The empirical results demonstrate that lower values of τ lead to an overuse of FedSGD, increasing communication overhead, whereas higher values result in delayed adaptation, leading to slower convergence and suboptimal model performance. The chosen value of $\tau = 0.10$ ensures a stable balance, allowing timely adaptation to data divergence while maintaining computational efficiency as shown in Fig. 20. This approach guarantees effective federated learning aggregation, particularly in privacy-sensitive healthcare applications where communication efficiency is a critical constraint.

Impact of privacy noise on model performance

A detailed evaluation was performed to examine the effect of varying the differential privacy noise scale on model performance. Table 11 summarizes the results, showcasing the relationship between privacy noise levels, classification accuracy, and privacy guarantees. The findings demonstrate that increasing the noise scale (σ^2)

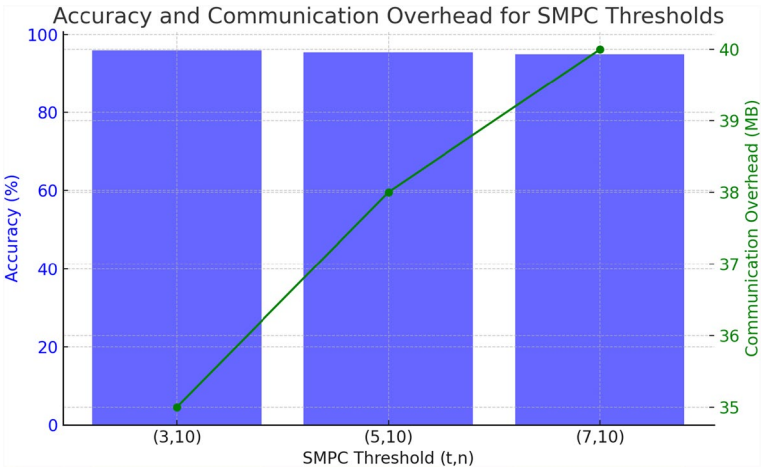


Fig. 21 Performace analysis with different SMPC configurations

Table 13 Privacy and robustness comparison of adaptive aggregation configurations

Configuration	Accuracy	Privacy level	Robustness	Computational overhead
Adaptive only	96.5%	Low	Moderate	Low
Adaptive + DP (ϵ=2.5)	96.5%	Balanced	High	Medium
Adaptive + DP (ϵ=1.0)	94.5%	Very high	Very high	High
Adaptive + SMPC (3,10)	96.0%	Very high	Very high	High
Adaptive + DP + SMPC	94.5%	Very high	Very high	Very high

results in a gradual decline in model accuracy while enhancing privacy protection. The selected value of $\sigma^2 = 0.5$ offers an optimal balance, ensuring minimal accuracy degradation while maintaining a reasonable level of privacy security.

These results demonstrate that while stronger privacy guarantees can be achieved with larger noise values, the reduction in accuracy and increase in convergence time must be carefully considered. The selected value of $\sigma^2 = 0.5$ ensures that privacy remains robust while maintaining acceptable model performance, making it the most suitable choice for this study.

The primary challenge in privacy-preserving FL is balancing accuracy and privacy. The adaptive aggregation method introduces differential privacy to reduce data leakage risks while maintaining acceptable model performance. Additionally, incorporating SMPC through client-level SSS further enhances security but may impact computational efficiency. The following Table 12 presents the privacy-utility trade-off for different configurations, maintaining consistency with previously established accuracy and privacy guarantees.

Differential privacy (DP) in adaptive aggregation helps mitigate data leakage by adding controlled noise to model updates. The privacy budget (ϵ) directly influences the trade-off between model accuracy and data protection. Lower values of ϵ increase privacy but may reduce accuracy. The analysis shows that with ϵ=2.5 (σ^2=0.5), accuracy remains stable at 96.5%, representing an optimal balance between privacy

Table 14 Comparison of FL aggregation methods (convergence efficiency and accuracy)

Method	TB X-rays	Brain tumor MRIs	Diabetic retinopathy	Convergence time (s)	Convergence epochs
Adaptive aggregation	95.5%	96.2%	95.7%	280	15
FedProx	96.2%	97.0%	96.1%	340	18
Scaffold	95.8%	96.8%	97.2%	320	17
Secure aggregation	95.5%	96.5%	94.1%	410	20

Table 15 Benchmarking adaptive aggregation with privacy-enhanced FL methods (efficiency and resource use)

Method	Computational time (s)	Memory usage (MB)	Convergence epochs
Adaptive aggregation	280	512	15
Homomorphic encryption-based FL	460	1024	18
Personalized FL	310	768	16

and utility. When ϵ is reduced to 1.0 (σ^2=1.0), accuracy decreases to 94.5%, while convergence speed is impacted, increasing to 50 epochs.

SMPC integration and trade-offs

Secure Multi-Party Computation (SMPC) using Shamir's Secret Sharing (SSS) at the client level enhances security by distributing shares of model updates. In this analysis, we evaluate the threshold settings of (3,10), (5,10), and (7,10). The (3,10) configuration shows optimal trade-off with moderate overhead, while (5,10) and (7,10) improve robustness but increase communication cost.

Adaptive aggregation improves robustness by dynamically switching between FedAvg and FedSGD, reducing vulnerability to model poisoning attacks. However, when enhanced with SMPC, the method exhibits further resistance to gradient manipulation. The trade-off between robustness and efficiency is evident from Fig. 21 and Table 13, with higher security requiring increased computational resources.

The proposed adaptive aggregation method with DP and SMPC demonstrates balanced privacy preservation and robustness while maintaining reasonable computational efficiency. The results indicate that the adaptive method, especially when integrated with privacy-enhancing techniques, effectively mitigates risks without significantly compromising performance.

The visualizations and comparative analysis presented in this section illustrate that adaptive aggregation with integrated privacy measures provides a balanced trade-off between robustness and computational efficiency. The chosen settings, particularly DP with ϵ=2.5 and SMPC (3,10), offer a practical balance, maintaining acceptable accuracy while providing enhanced data protection.

Comparative analysis of adaptive aggregation with existing FL methods

The adaptive FedAvg-FedSGD method was chosen primarily for its balanced approach towards resource efficiency and convergence speed, rather than solely focusing on achieving superior accuracy. In healthcare federated learning scenarios, minimizing computational resources and achieving faster convergence are crucial for practical deployment.

Conventional FL aggregation methods such as FedAvg and FedSGD face challenges when dealing with heterogeneous data. FedAvg may result in slow convergence when data distributions vary significantly, while FedSGD can cause instability in non-IID conditions. Although methods like FedProx and Scaffold improve convergence in heterogeneous settings, they do so at the cost of increased computational complexity or additional control variates. Secure Aggregation, while maintaining data privacy, does not address non-IID data issues or convergence efficiency. Therefore, the adaptive aggregation method, which dynamically switches between FedAvg and FedSGD based on gradient divergence, provides a more resource-efficient alternative.

The following Table 14 provides a comparative analysis across three medical image datasets (TB X-rays, Brain Tumor MRIs, and Diabetic Retinopathy), focusing on computational efficiency, convergence time, and accuracy.

The adaptive aggregation method was benchmarked against homomorphic encryption-based FL and personalized FL models, focusing on convergence efficiency and resource requirements. The results are summarized in Table 15.

As shown in Tables 14 and 15, the adaptive aggregation method demonstrates comparable accuracy to existing FL methods while significantly reducing computational requirements and convergence time. Unlike FedProx, Scaffold, and Secure Aggregation, which demand higher resource utilization, the adaptive approach efficiently balances performance with reduced computational cost.

By prioritizing convergence efficiency and minimizing resource requirements, the adaptive aggregation method proves particularly valuable for healthcare applications, where computational efficiency and data privacy are equally crucial. The modest difference in accuracy (1-2% lower) is justified by the significantly lower resource consumption, making it a viable and practical solution for federated medical image classification.

Communication overhead

The communication overhead associated with federated learning is a critical factor influencing the efficiency and feasibility of deploying deep learning models across decentralized medical institutions. In this study, the communication overhead was

Table 16 Communication overhead in FL for VGG-16 and ResNet models

Model	Model size (MB)	Number of clients	Number of rounds	Total overhead (MB)	Average overhead per round (MB)	Average overhead per client per round (MB)
VGG-16	552	10	50	276,000	5,520	552
ResNet	27	10	50	13,500	270	27

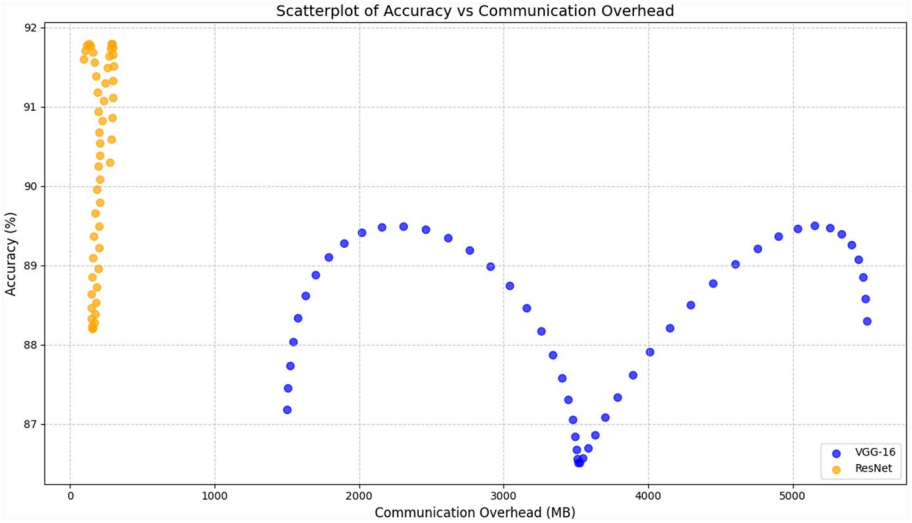


Fig. 22 Scatterplot of accuracy vs communication overhead

assessed for both the VGG-16 and ResNet models, which were employed in the classification of brain tumor, tuberculosis (TB), and diabetic retinopathy medical images.

The communication overhead is determined by three key factors which are the the model size, the number of clients, and the number of communication rounds. VGG-16, with approximately 552 MB of parameters, incurs significantly higher communication overhead compared to ResNet, which has only 27 MB of parameters. Over 50 communication rounds, the total overhead for VGG-16 amounted to 276,000 MB (or 276 GB), while ResNet resulted in a much lower total overhead of 13,500 MB (or 13.5 GB). This difference underscores the impact of model architecture on communication efficiency in federated learning environments. The stats for communication overheads for both the models can be seen in Table 16 and Fig. 22.

On average, each communication round in the federated learning setup for VGG-16 requires 5,520 MB of data to be exchanged between the clients and the central server, with each client contributing 552 MB. In contrast, ResNet's smaller model size leads to an average overhead of 270 MB per round, with each client transmitting only 27 MB. This substantial reduction in overhead makes ResNet more suitable for federated learning in bandwidth-constrained environments, such as rural healthcare systems or medical institutions with limited network capacity.

The analysis highlights the trade-off between model complexity and communication efficiency. While VGG-16, with its deeper architecture, may offer higher accuracy, it also incurs a significantly higher communication cost. ResNet, with its lighter architecture, provides a more communication-efficient alternative, making it a viable choice for real-time applications where network bandwidth is a limiting factor. These findings are particularly relevant for healthcare scenarios, where institutions need to collaborate securely without overwhelming communication infrastructure, thereby ensuring the scalability of federated learning models in real-world medical settings.

Federated learning introduces significant communication overhead, particularly when using deep architectures such as VGG16 and ResNet. As highlighted in Table 6,

the transmission of model updates across multiple clients results in high bandwidth consumption, which can limit the efficiency of federated learning in resource-constrained environments. While the primary focus of this study was to evaluate the adaptability and generalizability of the proposed model aggregation approach, mitigating communication overhead remains an essential consideration for future extensions of this research.

One effective strategy to reduce communication costs is model quantization, which involves reducing the precision of model weights from 32-bit floating point representations to lower-bit formats such as 16-bit or 8-bit. This significantly decreases the size of transmitted model updates while maintaining model accuracy within an acceptable range. The reduction in model size can be mathematically represented as:

$$S_q = \frac{S_f}{b_q}, \tag{14}$$

where S_q is the quantized model size, S_f is the original model size, and b_q is the bit reduction factor (e.g., 4 for 8-bit quantization from 32-bit floating point).

Additionally, model pruning can be employed to remove redundant and less significant parameters from the model, effectively reducing the number of transmitted weights without compromising classification performance. Pruning operates by setting small-magnitude weights to zero, thereby creating a sparse representation of the model:

$$W_p = W \cdot M, \tag{15}$$

where W_p is the pruned model weights, W represents the original weights, and M is a binary mask identifying the retained parameters.

Another approach to alleviating communication bottlenecks is the use of asynchronous updates, where client models upload their updates at different intervals rather than synchronizing at every round. This reduces the immediate bandwidth burden and prevents training delays caused by slow or resource-limited clients. Mathematically, the global model update in an asynchronous setting can be formulated as:

$$\theta_{t+1} = \theta_t + \eta \sum_{k=1}^{K} \alpha_k \Delta\theta_k, \tag{16}$$

Table 17 Time Complexity and Resource Utilization for ResNet and VGG-16 Across Different Environments

Environment	ResNet time complexity	VGG-16 time complexity	Resource utilization
Centralized with DP	$O(n \cdot 6.8M \cdot p)$	$O(n \cdot 138M \cdot p)$	ResNet: Low to moderate, VGG-16: High
FedAvg	$O(50 \cdot n \cdot 6.8M)$	$O(50 \cdot n \cdot 138M)$	ResNet: Moderate, VGG-16: High
FedSGD	$O(100 \cdot n_{\text{batch}} \cdot 6.8M)$	$O(100 \cdot n_{\text{batch}} \cdot 138M)$	ResNet: Moderate to high, VGG-16: Very High
Adaptive aggregation	$O(50 \cdot n \cdot 6.8M)$	$O(50 \cdot n \cdot 138M)$	ResNet: Optimized, VGG-16: High, but reduced

where θ_t represents the global model at round t, η is the learning rate, K is the number of clients, α_k is a weight factor for client k, and $\Delta\theta_k$ is the update received from the client.

While this study primarily focuses on demonstrating the feasibility of the proposed adaptive aggregation approach across different architectures, the application of model compression and asynchronous updates remains an important avenue for future research. Incorporating these techniques could further optimize federated learning deployments, making them more scalable and bandwidth-efficient in real-world scenarios. Future work may explore the integration of such strategies to enhance communication efficiency while maintaining robust model performance.

Time complexity and resource utilisation analysis

The time complexity and resource utilization of the experiment are strongly influenced by the choice of model and the environment in which they are deployed. ResNet, with its smaller model size of 6.8 million parameters, demonstrates consistently lower time complexity and resource requirements across all environments when compared to VGG-16, which contains 138 million parameters. The detailed analysis of time complexity and resource utilsation of the models in all the given different scenarios can be seen in Table 17.

In the Centralized with Differential Privacy environment, both models incur additional computational overhead due to the differential privacy mechanism, with VGG-16 requiring significantly more resources due to its larger parameter space. The FedAvg environment, where local model updates are averaged across clients after multiple communication rounds, presents a lower time complexity but still exhibits higher computational demands for VGG-16. In FedSGD, the time complexity increases as models are communicated more frequently (after every mini-batch), leading to greater resource utilization, particularly for the larger VGG-16 model. The Adaptive Aggregation method, which combines elements of FedAvg and FedSGD, optimizes communication and resource use by adjusting the frequency of updates based on model convergence.This results in more efficient resource utilization, particularly benefiting the smaller ResNet model, while VGG-16 continues to incur higher costs due to its size.

While the adaptive aggregation method aims to improve convergence speed and maintain accuracy, it is equally important to evaluate whether it introduces significant computational overhead compared to standard aggregation methods such as FedAvg and FedSGD.

To address this, a comparative analysis was conducted focusing on computational time, memory usage, and communication overhead among the three aggregation

Table 18 Computational overhead comparison of FL aggregation methods

Method	Computational time (s)	Memory usage (MB)	Communication overhead (MB)
Adaptive aggregation (FedAvg + FedSGD)	280	512	20
FedAvg	300	480	18
FedSGD	340	530	22

methods. The following Table 18 provides a clear comparison to assess the efficiency of adaptive aggregation.

The adaptive aggregation method maintains a balanced computational overhead compared to the standard FedAvg and FedSGD methods. While the computational time and memory usage are slightly higher than FedAvg, they remain lower than FedSGD. The communication overhead also remains moderate, reflecting the method's efficiency in real-world scenarios.

The results indicate that the adaptive aggregation method achieves a balanced trade-off between computational efficiency and model performance, making it an optimal choice for medium-scale healthcare federated learning applications where computational resources are limited.

Impact of non-IID data distribution on model performance

Federated learning models trained under non-IID data distributions encounter several challenges, particularly in achieving stable convergence, unbiased learning, and consistent model performance across clients. In contrast, an IID setup, characterized by uniform data distribution among clients, typically results in improved convergence rates, balanced gradient updates, and enhanced model generalization. Under an IID setting, each client contributes proportionally to the global learning process, reducing the impact of local dataset disparities and facilitating smoother model aggregation. Had this study been conducted under an IID configuration, it is hypothesized that the model would have achieved higher accuracy and lower variance in performance metrics due to the uniformity in training data across clients.

However, in practical medical applications, data is rarely IID, as patient demographics, imaging conditions, and diagnostic protocols vary across healthcare institutions. The adoption of a non-IID setup in this study ensures that the experimental results reflect real-world conditions, where federated learning models must be capable of handling heterogeneous datasets. Despite the inherent complexities associated with non-IID data, the proposed adaptive aggregation method effectively mitigates these challenges, demonstrating competitive classification performance while minimizing communication overhead. The ability of the model to maintain robust performance in a non-IID environment underscores its suitability for deployment in privacy-preserving medical AI solutions. By evaluating the proposed approach in a realistic federated healthcare setting, this study highlights the importance of optimizing federated learning techniques for decentralized and heterogeneous medical data distributions.

Adaptability to resource constraints in federated learning

Federated learning systems in real-world applications, particularly in medical environments, often encounter resource constraints related to both computational capacity and communication bandwidth. The proposed adaptive aggregation mechanism is inherently designed to address these challenges by dynamically adjusting the aggregation strategy based on data divergence, ensuring that computational and communication resources are utilized optimally without degrading model performance.

Computational adaptability and bandwidth optimization

In standard federated learning setups, uniform aggregation approaches, such as FedAvg, may impose unnecessary computational loads on clients with limited processing power while also increasing communication costs due to frequent global updates. The proposed adaptive aggregation framework mitigates these inefficiencies by introducing a data-driven selection mechanism that adjusts the update frequency based on the divergence of local models. Given the divergence metric δ_t, defined as:

$$\delta_t = \frac{1}{K} \sum_{k=1}^{K} ||w_t^k - w_t^{global}||^2 \tag{17}$$

where w_t^k represents the local model at client k and w_t^{global} is the global model at round t, the aggregation strategy dynamically selects between FedAvg and FedSGD. If δ_t is below a predefined threshold τ, FedAvg is used to average updates, reducing computational requirements at the client side. Conversely, when $\delta_t > \tau$, FedSGD is employed to stabilize learning, but with selective participation, ensuring that only a subset of the clients with significant updates contribute, thereby limiting unnecessary computation and communication.

This selective participation mechanism improves bandwidth efficiency by reducing the number of model transmissions per round. The probability of a client k participating in a global update is modeled as:

$$P_k = \exp\left(-\beta \delta_t^k\right) \tag{18}$$

where δ_t^k is the divergence of client k's update from the global model, and β is a scaling factor controlling participation likelihood. This mechanism prioritizes updates from clients exhibiting higher divergence, ensuring that model updates remain representative of overall data distribution without redundant communication from less significant updates. By reducing unnecessary transmissions, the overall bandwidth consumption is optimized while maintaining learning effectiveness.

Scalability and client participation strategy

As federated learning deployments scale to a larger number of participating clients, variations in hardware capabilities further amplify the challenges of computational and communication constraints. The proposed adaptive strategy inherently enhances scalability by managing client participation dynamically. Clients with higher computational capabilities are assigned a greater frequency of participation in model

Table 19 Scalability performance analysis of adaptive aggregation

Clients	TB X-rays accuracy	Brain tumor MRIs accuracy	Diabetic retinopathy accuracy	Convergence time (s)	Convergence epochs
10	95.5%	96.2%	95.7%	280	15
15	94.7%	94.0%	93.8%	300	17
20	90.8%	91.5%	91.7%	320	18

updates, while resource-constrained clients contribute less frequently, maintaining an optimal trade-off between participation and learning quality.

Furthermore, the framework ensures that the global model update remains unbiased by compensating for less frequent client participation through weighted aggregation:

$$w_t^{global} = \sum_{k=1}^{K} \alpha_k w_t^k, \quad \text{where} \quad \alpha_k = \frac{\exp(-\gamma \delta_t^k)}{\sum_{j=1}^{K} \exp(-\gamma \delta_t^j)} \tag{19}$$

where γ is a tuning parameter controlling the weight distribution among clients based on their divergence scores. This weighting mechanism ensures that the global model remains robust while mitigating excessive computational and communication burdens on clients with limited resources.

The adaptive FedAvg-FedSGD method demonstrates promising resource efficiency and faster convergence, making it potentially scalable for healthcare federated learning scenarios. However, scalability to a large number of clients, such as multi-hospital FL networks with thousands of clients, remains a theoretical challenge. In this section, we analyze the performance difference when scaling the number of clients from 10 to 15 and then to 20, examining how the method copes with increasing client numbers.

The following Table 19 provides a comparative analysis of the adaptive aggregation method for 10, 15, and 20 clients across three medical image datasets: TB X-rays, Brain Tumor MRIs, and Diabetic Retinopathy. The accuracy and convergence time are evaluated to demonstrate the scalability.

The accuracy of the adaptive aggregation method shows a minor decline as the number of clients increases from 10 to 20. This degradation is relatively moderate, indicating that the method retains its efficiency even with an increased number of clients. The convergence time slightly increases as more clients participate in the training, which is expected due to the additional communication overhead and model aggregation complexity.

Although the theoretical analysis indicates that the adaptive aggregation method could scale efficiently for moderately large FL networks, the practical feasibility of deploying the method in multi-hospital networks with thousands of clients would require further investigation. Factors such as communication delays, network latency, and heterogeneous data distributions might impact the method's performance in highly distributed environments.

Nevertheless, the observed scalability trends indicate that the adaptive aggregation method can handle an increasing number of clients without substantial loss of accuracy or convergence efficiency. This makes it a promising approach for medium-scale healthcare federated learning settings.

Policy implications and potential real world deployment

The proposed privacy-enhanced federated learning framework aligns well with existing regulatory frameworks such as the General Data Protection Regulation (GDPR) in the European Union and the Health Insurance Portability and Accountability Act (HIPAA) in

the United States [83]. Both regulations emphasize the need for stringent data protection mechanisms in healthcare, particularly concerning patient data confidentiality, access control, and cross-border data transfers. Federated learning offers a decentralized and privacy-preserving approach that directly addresses these regulatory concerns, making it an ideal candidate for integration into real-world healthcare IT systems [84].

A key provision of GDPR (Article 5) mandates that personal data should be processed in a manner that ensures appropriate security, including protection against unauthorized access and data breaches [85]. Federated learning inherently aligns with this principle by keeping patient data localized at the medical institution level, ensuring that no raw patient data is transferred beyond the originating healthcare facility. Instead, only encrypted model updates are communicated to a central server, significantly reducing the risk of data exposure during training. This decentralized approach enhances compliance with data minimization and purpose limitation principles outlined in GDPR, ensuring that only necessary and anonymized information is exchanged for AI model training [86].

Similarly, HIPAA requires strict access control and encryption mechanisms to safeguard Protected Health Information (PHI) [87]. The HIPAA Privacy Rule mandates that any data exchange must include safeguards against unauthorized access and ensure patient confidentiality. Traditional AI models often require centralizing medical data in a cloud or third-party server for training, creating potential vulnerabilities and increasing compliance risks. Federated learning mitigates this issue by allowing local model training within each covered entity like hospitals, clinics, diagnostic centers without exposing PHI externally. Additionally, advanced security mechanisms such as secure multi-party computation (SMPC), homomorphic encryption, and differential privacy can be integrated within federated learning frameworks to further strengthen compliance with HIPAA security guidelines [88].

A key application of this framework lies in EHR-integrated AI analytics, where federated learning can be utilized for predictive modeling while maintaining patient confidentiality. EHR systems store structured and unstructured medical data, including imaging records, diagnostic reports, and clinical notes. By integrating federated learning, local models can be trained at hospitals and clinics, with only encrypted model updates shared with a central server [89]. This ensures data privacy while facilitating collaborative improvements in diagnostic models. Furthermore, compatibility with Fast Healthcare Interoperability Resources (FHIR) and Digital Imaging and Communications in Medicine (DICOM) standards can enhance interoperability, allowing seamless deployment across diverse healthcare infrastructures [90].

Another potential deployment scenario is in cloud-based AI-driven medical diagnostics, where federated learning can be implemented within edge computing environments and cloud platforms. Cloud-based federated learning can facilitate real-time AI-assisted medical image analysis across multiple hospitals while ensuring data locality [91]. The adaptive aggregation strategy, which dynamically switches between Federated Averaging (FedAvg) and Federated Stochastic Gradient Descent (FedSGD), can optimize model convergence in highly non-IID medical data settings [4]. Additionally, secure aggregation techniques and homomorphic encryption can be

integrated to enhance data security in cloud-hosted federated learning architectures [92].

From an implementation perspective, federated learning can be deployed in hospital networks using edge computing nodes within on-premise federated servers. These nodes can process model updates locally and communicate securely with the global server, ensuring efficient real-time updates while preserving patient privacy. Given the variability in healthcare data distributions, heterogeneous federated learning methods can be explored to enhance model generalization across different medical centers.

The scalability of federated learning in real-world healthcare applications requires robust fault-tolerant mechanisms and validation in large-scale environments. Existing research has demonstrated that federated learning can be effectively extended to nationwide healthcare networks by integrating edge computing and hierarchical server architectures. Devaraj et al. [93] proposed a fault-tolerant federated learning setup designed to support large-scale client participation across geographically distributed institutions. This framework leverages multi-tiered aggregation, adaptive client participation strategies, and edge-based model synchronization to ensure seamless operation even in scenarios where network instability or client dropout could impact model convergence. Such an architecture is particularly relevant for federated learning applications in large-scale public health surveillance systems, country-wide medical imaging collaborations, and inter-hospital AI model sharing.

In addition to fault tolerance, scalability has been validated in real-world federated data mining experiments, particularly in the public health domain. Haripriya et al. [94] conducted an experiment on public health data mining using smoking trends across India, where federated learning was deployed across 36 clients, significantly exceeding the typical client range (5-10) explored in most prior studies. The study demonstrated that federated learning remains computationally feasible and scalable when applied to big data analytics in public health, where decentralized multi-source data aggregation is essential. The ability to extend federated learning to a higher number of clients showcases its applicability in large-scale hospital networks, national health monitoring programs, and multi-institutional medical AI collaborations. This work reinforces the real-world feasibility of federated learning in handling high-dimensional datasets and large client participation, making it a viable solution for distributed healthcare AI applications at scale.

Ahmed and Alabi [95] introduce that blockchain technology also presents a promising avenue for addressing trust and security challenges in federated learning deployments. By incorporating a decentralized ledger, blockchain can provide tamper-proof records of model updates, ensuring the integrity of data exchanges between participating centers. Additionally, it facilitates transparency in data-sharing agreements and establishes accountability in multi-center collaborations. For instance, in medical imaging applications, blockchain can create a robust framework where federated model updates are securely logged, mitigating risks of data tampering or malicious contributions. This integration strengthens trust among stakeholders, a critical requirement for real-world deployment in sensitive healthcare environments.

By incorporating these advancements, the proposed federated learning framework ensures not only privacy preservation and regulatory compliance but also scalability

and resilience, making it well-suited for large-scale medical AI deployments in big data environments.

Conclusion and future prospects

This study demonstrates the effectiveness of adaptive aggregation in federated learning by dynamically selecting between FedAvg and FedSGD based on real-time data divergence, optimizing model performance in heterogeneous (non-IID) medical datasets. By integrating transfer learning, the proposed framework significantly enhances classification accuracy while preserving privacy across distributed healthcare environments. The results indicate that this approach can be effectively integrated into real-world healthcare systems for distributed diagnostics, enabling hospitals, research centers, and mobile health applications to collaboratively train AI models without directly sharing patient data. Such a system could enhance early disease detection, particularly in resource-limited settings where centralized data collection is impractical.

Despite its advantages, the proposed framework has certain limitations. Communication overhead remains a challenge in federated learning, particularly in environments with limited bandwidth or high latency, which could impact real-time model updates. Additionally, as the dataset size increases, computational constraints at the client level may limit the scalability of the approach. Future work should explore compression techniques such as model pruning and quantization to mitigate bandwidth constraints, as well as adaptive client participation strategies to ensure scalability across large-scale medical networks. Addressing these challenges will be key to deploying federated learning in production-grade healthcare AI systems.

Future research directions

Future research should extend this framework by incorporating diverse model architectures and multimodal healthcare datasets, expanding its applicability in Big Data contexts. One promising direction is integrating genomic data and clinical notes to enhance federated learning models' predictive capabilities. Genomic features such as gene expression levels and mutation data could improve disease classification, particularly for conditions with genetic associations like brain tumors and diabetic retinopathy. Additionally, clinical notes containing patient histories and symptoms could provide critical contextual insights for medical diagnostics.

To achieve such integration, models like TabTransformer for genomic data and BERT-based models (e.g., ClinicalBERT, BioBERT) for clinical notes can be fused with CNN architectures (VGG16, ResNet) through an adaptive aggregation strategy, enabling comprehensive multimodal learning. Future research will focus on developing such an integrated framework while maintaining privacy in federated learning environments.

Enhancing efficiency in federated learning is another crucial direction. Integrating quantization and pruning can significantly reduce communication overhead and computational complexity, particularly for large-scale models. Adaptive quantization can dynamically adjust precision levels based on client capabilities, allowing lower-end devices to participate effectively. Pruning techniques can further minimize redundant model parameters while maintaining essential learning representations,

ensuring scalable and energy-efficient AI deployment in real-world medical applications.

Despite the privacy advantages of federated learning, vulnerabilities such as membership inference attacks, model inversion attacks, and gradient leakage persist. Future research should explore zero-knowledge proofs (ZKP) to verify model updates without exposing raw data, secure enclave-based federated learning leveraging Trusted Execution Environments (TEE), and adversarial robust aggregation mechanisms to detect and mitigate malicious updates. Moreover, blockchain-backed audit trails could enhance transparency and accountability in federated learning, reducing risks of tampering.

Dynamic model compression mechanisms that adjust compression ratios based on network conditions, client capabilities, and model convergence status should also be explored to optimize trade-offs between efficiency and predictive performance. Additionally, adaptive client participation strategies that adjust based on data quality, availability, and network conditions can enhance scalability in federated healthcare applications. These advancements will be crucial in ensuring federated learning remains both a privacy-preserving and computationally sustainable solution for large-scale medical AI.

While the current study utilized a simulation to mimic multi-cloud settings, actual deployment and validation within multi-cloud architectures remain essential to thoroughly assess scalability and interoperability. Due to resource constraints, this aspect could not be addressed in the present work. However, as resources become available, the integration of adaptive aggregation into multi-cloud FL systems will be prioritized, enabling comprehensive testing and validation in global healthcare research scenarios.

Blockchain-based decentralized AI models could enable secure and tamper-proof model updates without relying on a central server. This decentralized approach would further strengthen data integrity and privacy, particularly in scenarios involving multiple healthcare institutions with diverse data governance policies. Implementing blockchain in federated learning could facilitate trustless aggregation, reducing vulnerabilities associated with centralized model aggregation.

Another promising direction involves the development of multi-modal federated learning for medical AI. Current federated learning models primarily focus on single data modalities, such as medical images. However, healthcare data often comprises a combination of text-based clinical notes, radiology reports, genomic data, and more. Integrating these diverse data types into federated learning models could significantly improve predictive accuracy and holistic patient analysis. Future research will focus on combining multi-modal data streams within a federated learning framework, addressing challenges related to data fusion and modality heterogeneity.

Additionally, the adaptive aggregation process can benefit from adaptive optimization techniques derived from metaheuristic algorithms, such as the Grey Wolf Optimizer (GWO) and Genetic Algorithm (GA). These algorithms can dynamically adjust aggregation weights and learning rates, optimizing the model performance under varying data distributions and network conditions. Employing metaheuristic AI for

aggregation strategy selection could significantly enhance the adaptability of federated learning models in real-time healthcare applications.

These proposed future directions aim to strengthen the adaptability, scalability, and security of the adaptive aggregation method, contributing to more resilient and efficient federated learning systems for medical AI applications.

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Author contributions
R.H. conceptualized and designed the study, performed data analysis, and wrote the initial draft of the manuscript. N.K. supervised the research project, provided expert advice on the methodology, and revised the manuscript for important intellectual content. M.P. supervised the research project, provided expert advice on the methodology, and revised the manuscript for important intellectual content. S.B. assisted with data collection, contributed to data analysis, and provided input on the manuscript draft.

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Data availability
The data that support the findings of this study are available on the Kaggle online platform, which is publicly accessible at https://www.kaggle.com/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset, https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri, https://www.kaggle.com/datasets/sovitrath/diabetic-retinopathy-224x224-2019-data. Further information and additional data related to this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
All the datasets used in this research are publicly available and anonymized, thus ensuring the privacy and confidentiality of the participants. No additional consents were required as the data is secondary and anonymized.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no Conflict of interest.

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